

LOC	OBJECT CODE	ADDR1	ADDR2	STMT
				2 *****
				3 *
				4 * EXPERIMENTAL pending further PoP definition
				5 *
				6 * Zvector E6 instruction tests for VRR-c encoded:
				7 *
				8 * E675 VCRNF - VECTOR FP CONVERT AND ROUND TO NNP
				9 *
				10 * and partial testing of
				11 *
				12 * E656 VCLFNH - VECTOR FP CONVERT AND LENGTHEN FROM NNP HIGH
				13 * E65E VCLFNL - VECTOR FP CONVERT AND LENGTHEN FROM NNP LOW
				14 *
				15 * during cross check tests for VCRNF
				16 *
				17 * James Wekel August 2024
				18 *****
				20 *****
				21 *
				22 * basic instruction tests
				23 *
				24 *****
				25 * This program tests EXPERIMENTAL functioning of the z/arch E6 VRR-c
				26 * Neural-network-processing-assist facility vector instructions.
				27 * These test are EXPERIMENTAL pending further PoP definition of
				28 * NNP-data-type-1.
				29 *
				30 * If requested and if VXC == 0 after test instruction execution,
				31 * a cross check test is performed. A cross check uses the result
				32 * of the instruction test to recreate the test source.
				33 *
				34 * Exceptions (including trapable IEEE exceptions) are not tested.
				35 *
				36 * PLEASE NOTE that the tests are very SIMPLE TESTS designed to catch
				37 * obvious coding errors. None of the tests are thorough. They are
				38 * NOT designed to test all aspects of any of the instructions.
				39 *
				40 *****
				41 *
				42 * *Testcase zvector-e6-21-VCRNF
				43 * *
				44 * * EXPERIMENTAL pending further PoP definition
				45 * *
				46 * * Zvector E6 instruction tests for VRR-c encoded:
				47 * *
				48 * * E675 VCRNF - VECTOR FP CONVERT AND ROUND TO NNP
				49 * *
				50 * * # -----
				51 * * # This tests only the basic function of the instruction.
				52 * * # Exceptions are NOT tested.
				53 * * # -----
				54 * *
				55 * mainsize 2
				56 * numcpu 1

```

57 *      sysclear
58 *      archlvl      z/Arch
59 *
60 *      loadcore      "$(testpath)/zvector-e6-21-VCRNF.core" 0x0
61 *
62 *      diag8cmd      enable      # (needed for messages to Hercules console)
63 *      runtest 5
64 *      diag8cmd      disable     # (reset back to default)
65 *
66 *      *Done
67 *
68 ****

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LOC	OBJECT CODE	ADDR1	ADDR2	STMT
				70 *****
				71 * FCHECK Macro - Is a Facility Bit set?
				72 *
				73 * If the facility bit is NOT set, an message is issued and
				74 * the test is skipped.
				75 *
				76 * Fcheck uses R0, R1 and R2
				77 *
				78 * eg. FCHECK 134,'vector-packed-decimal'
				79 *****
				80 MACRO
				81 FCHECK &BITNO,&NOTSETMSG
				82 .* &BITNO : facility bit number to check
				83 .* &NOTSETMSG : 'facility name'
				84 LCLA &FBBYTE Facility bit in Byte
				85 LCLA &FBBIT Facility bit within Byte
				86
				87 LCLA &L(8)
				88 &L(1) SetA 128,64,32,16,8,4,2,1 bit positions within byte
				89
				90 &FBBYTE SETA &BITNO/8
				91 &FBBIT SETA &L((&BITNO-(&FBBYTE*8))+1)
				92 .* MNOTE 0,'checking Bit=&BITNO: FBBYTE=&FBBYTE, FBBIT=&FBBIT'
				93
				94 B X&SYSNDX
				95 * Fcheck data area
				96 * skip messgae
				97 SKT&SYSNDX DC C' Skipping tests: '
				98 DC C&NOTSETMSG
				99 DC C' facility (bit &BITNO) is not installed.'
				100 SKL&SYSNDX EQU *-SKT&SYSNDX
				101 * facility bits
				102 DS FD gap
				103 FB&SYSNDX DS 4FD
				104 DS FD gap
				105 *
				106 X&SYSNDX EQU *
				107 LA R0,((X&SYSNDX-FB&SYSNDX)/8)-1
				108 STFLE FB&SYSNDX get facility bits
				109
				110 XGR R0,R0
				111 IC R0,FB&SYSNDX+&FBBYTE get fbit byte
				112 N R0,=F'&FBBIT' is bit set?
				113 BNZ XC&SYSNDX
				114 *
				115 * facility bit not set, issue message and exit
				116 *
				117 LA R0,SKL&SYSNDX message length
				118 LA R1,SKT&SYSNDX message address
				119 BAL R2,MSG
				120
				121 B EOJ
				122 XC&SYSNDX EQU *
				123 MEND

LOC	OBJECT CODE	ADDR1	ADDR2	STMT
				144
				145 *****
				146 * The actual "ZVE6TST" program itself...
				147 *****
				148 *
				149 * Architecture Mode: z/Arch
				150 * Register Usage:
				151 *
				152 * R0 (work)
				153 * R1-4 (work)
				154 * R5 Testing control table - current test base
				155 * R6 (work)
				156 * R7 msg return
				157 * R8 First base register
				158 * R9 Second base register
				159 * R10 Third base register
				160 * R11 E6TEST call return
				161 * R12 E6TESTS register
				162 * R13 (work)
				163 * R14 Subroutine call
				164 * R15 Secondary Subroutine call or work
				165 *
				166 *****
00000200		00000200		168 USING BEGIN,R8 FIRST Base Register
00000200		00001200		169 USING BEGIN+4096,R9 SECOND Base Register
00000200		00002200		170 USING BEGIN+8192,R10 THIRD Base Register
				171
00000200	0580			172 BEGIN BALR R8,0 Initialize FIRST base register
00000202	0680			173 BCTR R8,0 Initialize FIRST base register
00000204	0680			174 BCTR R8,0 Initialize FIRST base register
				175
00000206	4190 8800		00000800	176 LA R9,2048(,R8) Initialize SECOND base register
0000020A	4190 9800		00000800	177 LA R9,2048(,R9) Initialize SECOND base register
				178
0000020E	41A0 9800		00000800	179 LA R10,2048(,R9) Initialize THIRD base register
00000212	41A0 A800		00000800	180 LA R10,2048(,R10) Initialize THIRD base register
				181
00000216	B600 8524		00000724	182 STCTL R0,R0,CTLR0 Store CR0 to enable AFP
0000021A	9604 8525		00000725	183 OI CTLR0+1,X'04' Turn on AFP bit
0000021E	9602 8525		00000725	184 OI CTLR0+1,X'02' Turn on Vector bit
00000222	B700 8524		00000724	185 LCTL R0,R0,CTLR0 Reload updated CR0
				186
				187 *****
				188 * Is Neural-network-processing-assist facility 2 installed (bit 165)
				189 *****
				190
00000226	47F0 80C0		000002C0	191 FCHECK 165,'Neural-network-processing-assist'
				192+ B X0001
				193+*
				194+* Fcheck data area
				195+SKT0001 DC C' Skipping tests: '
0000022A	40404040 40404040			196+ DC C'Neural-network-processing-assist'
00000244	D585A499 81936095			197+ DC C' facility (bit 165) is not installed.'
00000264	40868183 899389A3			198+SKL0001 EQU *-SKT0001
		0000005F 00000001		199+* facility bits

LOC	OBJECT	CODE	ADDR1	ADDR2	STMT	
					220 *****	
					221 *	
					222 *****	
					223	
000002E8	58C0	854C		0000074C	224 L R12,=A(E6TESTS)	get table of test addresses
			000002EC	00000001	225	
000002EC	5850	C000		00000000	226 NEXTE6 EQU *	
000002F0	1255				227 L R5,0(0,R12)	get test address
000002F2	4780	83F4		000005F4	228 LTR R5,R5	have a test?
					229 BZ ENDTEST	done?
					230	
000002F6			00000000		231 USING E6TEST,R5	
					232	
000002F6	4800	5004		00000004	233 LH R0,TNUM	save current test number
000002FA	5000	8E04		00001004	234 ST R0,TESTING	for easy reference
					235	
000002FE	58B0	5000		00000000	236 L R11,TSUB	get address of test routine
00000302	05BB				237 BALR R11,R11	do test
					238	
00000304	45F0	814E		0000034E	239 BAL R15,XCHECK	
					240 *	
					241 * validate FPC first	
					242 *	
00000308	D500	500A	5041	00000041	243 CLC FLG(1),FPC_R+1	expected FPC flags?
0000030E	4780	8116		00000316	244 BE CHECKVXC	
00000312	4570	8290		00000490	245 BAL R7,FPCMSG	no, issue failed message
					246	
00000316	D500	500B	5042	00000001	247 CHECKVXC EQU *	
				00000042	248 CLC VXC(1),FPC_R+2	expected VXC?
0000031C	4780	8124		00000324	249 BE CHECKRES	
00000320	4570	8302		00000502	250 BAL R7,VXCMSG	no, issue failed message
					251	
			00000324	00000001	252 CHECKRES EQU *	
					253	
					254 * then validate results, if not inexact	
					255	
00000324	B982	0011			256 XGR R1,R1	
00000328	4310	5041		00000041	257 IC R1,FPC_R+1	FPC flags
0000032C	5410	8550		00000750	258 N R1,=XL4'00000008'	check inexact flag
00000330	1211				259 LTR R1,R1	
00000332	4770	8146		00000346	260 BNZ DONEXT	
					261	
00000336	E310	501C	0014	0000001C	262 LGF R1,READDR	expected result address
0000033C	D50F	5028	1000	00000000	263 CLC V10OUTPUT,0(R1)	
00000342	4770	8374		00000574	264 BNE FAILMSG	no, issue failed message
					265	
			00000346	00000001	266 DONEXT EQU *	
00000346	41C0	C004		00000004	267 LA R12,4(0,R12)	next test address
0000034A	47F0	80EC		000002EC	268 B NEXTE6	

LOC	OBJECT CODE			ADDR1	ADDR2	STMT			
						270	*-----		
						271	* cross check that the result can be converted back to the source		
						272	* where there were NO exceptions (DXC/VXC = 0)		
						273	*-----		
				0000034E	00000001	274	XCHECK	EQU	*
0000034E	D207	5050	8530	00000050	00000730	275		MVC	SKIPXC,=CL8'SKIP XC '
00000354	D500	5007	855C	00000007	0000075C	276		CLC	XCSKIP,=CL1'S'
0000035A	478F	0000			00000000	277		BE	0(R15)
0000035E	9500	5042			00000042	278		CLI	FPC_R+2,0
00000362	477F	0000			00000000	279		BNZ	0(R15)
						280	* skip xcheck requested		
						281	* skip = S, so exit		
						282	* cross check depends on the instruction		
00000366	D207	5050	8538	00000050	00000738	283		MVC	SKIPXC,=CL8'
0000036C	D507	5010	8540	00000010	00000740	284		CLC	OPNAME,=CL8'VCRNF'
00000372	4780	8180			00000380	285		BE	XCVCRNF
						286			
00000376	D207	5050	8530	00000050	00000730	287		MVC	SKIPXC,=CL8'SKIP XC '
0000037C	07FF					288		BR	R15
						289	return from xcheck		
						290			
						291	* cross check: VCRNF - VECTOR FP CONVERT AND ROUND TO NNP		
						292	*-----		
00000380						293	XCVCRNF	DS	0F
00000380	B29D	852C			0000072C	294		LFPC	FPCINIT
						295	initialize FPC		
00000384	E700	5028	0806		00000028	296		VL	V16,V10OUTPUT
0000038A	E6F0	0000	2456			297		VCLFNH	V15,V16,2,0
00000390	B29C	5088			00000088	298		STFPC	FPC_XC1
00000394	E7F0	5060	000E		00000060	299		VST	V15,XCOUTPUT
						300	save 1st FPC		
0000039A	B29D	852C			0000072C	301		LFPC	FPCINIT
						302	initialize FPC		
0000039E	E700	5028	0806		00000028	303		VL	V16,V10OUTPUT
000003A4	E6F0	0000	245E			304		VCLFNL	V15,V16,2,0
000003AA	B29C	508C			0000008C	305		STFPC	FPC_XC2
000003AE	E7F0	5070	000E		00000070	306		VST	V15,XCOUTPUT+16
						307	save 2nd FPC		
000003B4	9500	508A			0000008A	308		CLI	FPC_XC1+2,0
000003B8	477F	0000			00000000	309		BNZ	0(R15)
						310	1st FPC		
000003BC	9500	508E			0000008E	311		CLI	FPC_XC2+2,0
000003C0	477F	0000			00000000	312		BNZ	0(R15)
						313	2nd FPC		
000003C4	E310	500C	0014		0000000C	314		LGF	R1,V2ADDR
000003CA	D51F	5060	1000		00000000	315		CLC	XCOUTPUT(32),0(R1)
000003D0	4770	81D6			000003D6	316		BNE	XCFAILMSG
000003D4	07FF					317		BR	R15
						318	return from xcheck		
						319			
000003D6						320	* xcheck failed message		
000003D6	5800	8554			00000754	321	XCFAILMSG	DS	0H
000003DA	5000	8E00			00001000	322		L	R0,=F'1'
						323		ST	R0,FAILED
						324	set failed test indicator		
000003DE	4820	5004			00000004	325		LH	R2,TNUM
							get test number and convert		

LOC	OBJECT	CODE	ADDR1	ADDR2	STMT			
000003E2	4E20	8F90		00001190	326	CVD	R2,DECNUM	
000003E6	D211	8F7A 8F64	0000117A	00001164	327	MVC	PRT3,EDIT	
000003EC	DE11	8F7A 8F90	0000117A	00001190	328	ED	PRT3,DECNUM	
000003F2	D202	8F23 8F87	00001123	00001187	329	MVC	XCPTNUM(3),PRT3+13	fill in message with test #
					330			
000003F8	D207	8F45 5010	00001145	00000010	331	MVC	XCPNAME,OPNAME	fill in message with instruction
					332			
000003FE	B982	0022			333	XGR	R2,R2	
00000402	4320	5008		00000008	334	IC	R2,M4	get m3 and convert
00000406	4E20	8F90		00001190	335	CVD	R2,DECNUM	
0000040A	D211	8F7A 8F64	0000117A	00001164	336	MVC	PRT3,EDIT	
00000410	DE11	8F7A 8F90	0000117A	00001190	337	ED	PRT3,DECNUM	
00000416	D201	8F56 8F88	00001156	00001188	338	MVC	XCPM4(2),PRT3+14	fill in message with m3 field
					339	*		
0000041C	B982	0022			340	XGR	R2,R2	
00000420	4320	5009		00000009	341	IC	R2,M5	get m4 and convert
00000424	4E20	8F90		00001190	342	CVD	R2,DECNUM	
00000428	D211	8F7A 8F64	0000117A	00001164	343	MVC	PRT3,EDIT	
0000042E	DE11	8F7A 8F90	0000117A	00001190	344	ED	PRT3,DECNUM	
00000434	D201	8F61 8F88	00001161	00001188	345	MVC	XCPM5(2),PRT3+14	fill in message with m4 field
					346			
0000043A	50F0	8288		00000488	347	ST	R15,XCR15	save r15
0000043E	4100	004E		0000004E	348	LA	R0,XCPLNG	message length
00000442	4110	8F16		00001116	349	LA	R1,XCPLINE	message address
00000446	45F0	8402		00000602	350	BAL	R15,RPTERROR	
0000044A	58F0	8288		00000488	351	L	R15,XCR15	
					352			
0000044E	5800	8554		00000754	353	L	R0,=F'1'	set failed test indicator
00000452	5000	8E00		00001000	354	ST	R0,FAILED	
00000456	07FF				355	BR	R15	return from xcheck
					356			
00000458					357	DS	0FD	
00000458	00000000	00000000			358	XCRESULT DS	XL16	
00000468	00000000	00000000			359	XCV1 DS	XL16	
00000478	00000000	00000000			360	XCV2 DS	XL16	
00000488	00000000	00000000			361	XCR15 DS	FD	

LOC	OBJECT CODE			ADDR1	ADDR2	STMT
						436 *****
						437 * result not as expected:
						438 * issue message with test number, instruction under test
						439 * and instruction m4
						440 *****
				00000574	00000001	441 FAILMSG EQU *
00000574	5800	8554			00000754	442 L R0,=F'1' set failed test indicator
00000578	5000	8E00			00001000	443 ST R0,FAILED
						444
0000057C	4820	5004			00000004	445 LH R2,TNUM get test number and convert
00000580	4E20	8F90			00001190	446 CVD R2,DECNUM
00000584	D211	8F7A 8F64		0000117A	00001164	447 MVC PRT3,EDIT
0000058A	DE11	8F7A 8F90		0000117A	00001190	448 ED PRT3,DECNUM
00000590	D202	8EDB 8F87		000010DB	00001187	449 MVC PRTNUM(3),PRT3+13 fill in message with test #
						450
00000596	D207	8EF6 5010		000010F6	00000010	451 MVC PRTNAME,OPNAME fill in message with instruction
						452 *
0000059C	B982	0022				453 XGR R2,R2
000005A0	4320	5008			00000008	454 IC R2,M4 get m3 and convert
000005A4	4E20	8F90			00001190	455 CVD R2,DECNUM
000005A8	D211	8F7A 8F64		0000117A	00001164	456 MVC PRT3,EDIT
000005AE	DE11	8F7A 8F90		0000117A	00001190	457 ED PRT3,DECNUM
000005B4	D201	8F07 8F88		00001107	00001188	458 MVC PRTM4(2),PRT3+14 fill in message with m3 field
						459 *
000005BA	B982	0022				460 XGR R2,R2
000005BE	4320	5009			00000009	461 IC R2,M5 get m4 and convert
000005C2	4E20	8F90			00001190	462 CVD R2,DECNUM
000005C6	D211	8F7A 8F64		0000117A	00001164	463 MVC PRT3,EDIT
000005CC	DE11	8F7A 8F90		0000117A	00001190	464 ED PRT3,DECNUM
000005D2	D201	8F13 8F88		00001113	00001188	465 MVC PRTM5(2),PRT3+14 fill in message with m4 field
						466 *
000005D8	4100	0048			00000048	467 LA R0,PRTLNG message length
000005DC	4110	8ECE			000010CE	468 LA R1,PRTLINE message address
000005E0	45F0	8402			00000602	469 BAL R15,RPTERROR
						471 *****
						472 * continue after a failed test
						473 *****
				000005E4	00000001	474 FAILCONT EQU *
000005E4	5800	8554			00000754	475 L R0,=F'1' set failed test indicator
000005E8	5000	8E00			00001000	476 ST R0,FAILED
						477
000005EC	41C0	C004			00000004	478 LA R12,4(0,R12) next test address
000005F0	47F0	80EC			000002EC	479 B NEXTE6
						481 *****
						482 * end of testing; set ending psw
						483 *****
				000005F4	00000001	484 ENDTEST EQU *
000005F4	5810	8E00			00001000	485 L R1,FAILED did a test fail?
000005F8	1211					486 LTR R1,R1
000005FA	4780	8508			00000708	487 BZ EOJ No, exit
000005FE	47F0	8520			00000720	488 B FAILTEST Yes, exit with BAD PSW
						489

LOC	OBJECT CODE	ADDR1	ADDR2	STMT	
				554	*****
				555	* Normal completion or Abnormal termination PSWs
				556	*****
000006F8	00020001 80000000			558	E0JPSW DC 0D'0',X'0002000180000000',AD(0)
00000708	B2B2 84F8		000006F8	560	E0J LPSWE E0JPSW Normal completion
00000710	00020001 80000000			562	FAILPSW DC 0D'0',X'0002000180000000',AD(X'BAD')
00000720	B2B2 8510		00000710	564	FAILTEST LPSWE FAILPSW Abnormal termination
				566	*****
				567	* Working Storage
				568	*****
00000724	00000000			570	CTLR0 DS F CR0
00000728	00000000			571	DS F
0000072C	00000000			572	FPCINIT DC XL4'00000000' FPC before test
00000730				574	
00000730	E2D2C9D7 40E7C340			575	LTORG , Literals pool
00000738	40404040 40404040			576	=CL8'SKIP XC '
00000740	E5C3D9D5 C6404040			577	=CL8'
00000748	00000004			578	=CL8'VCRNF'
0000074C	00002D04			579	=F'4'
00000750	00000008			580	=A(E6TESTS)
00000754	00000001			581	=XL4'00000008'
00000758	0000			582	=F'1'
0000075A	0064			583	=H'0'
0000075C	E2			584	=AL2(L'MSGMSG)
				585	=CL1'S'
				586	
				587	* some constants
				588	
	00000400 00000001			589	K EQU 1024 One KB
	00001000 00000001			590	PAGE EQU (4*K) Size of one page
	00010000 00000001			591	K64 EQU (64*K) 64 KB
	00100000 00000001			592	MB EQU (K*K) 1 MB
				593	

LOC	OBJECT CODE	ADDR1	ADDR2	STMT
				598 *=====
				599 *
				600 * NOTE: start data on an address that is easy to display
				601 * within Hercules
				602 *
				603 *=====
				604
0000075D		0000075D	00001000	605
00001000	00000000			606 FAILED DC F'0' some test failed?
00001004	00000000			607 TESTING DC F'0' current test #
				609
				610 *****
				611 * TEST failed : FPC message
				612 *****
				613 *
				614 * failed message and associated editing
				615 *
00001008	40404040	4040E385		616 FPCPLINE DC C' Test # '
00001015	A7A7A7			617 FPCPNUM DC C'xxx'
00001018	40A69996	958740C6		618 DC c' wrong FPC flags for instruction '
00001039	A7A7A7A7	A7A7A7A7		619 FPCPNAME DC CL8'xxxxxxxx'
00001041	4085A797	8583A385		620 DC C' expected: flags='
00001052	A7A7A7			621 FPCPEXP DC C'xxx'
00001055	6B			622 DC C','
00001056	40998583	8589A585		623 DC C' received: flags='
00001067	A7A7A7			624 FPCPGOT DC C'xxx'
0000106A	4B			625 DC C'.'
		00000063	00000001	626 FPCPLNG EQU *-FPCPLINE
				628
				629 *****
				630 * TEST failed : VXC message
				631 *****
				632 *
				633 * failed message and associated editing
				634 *
0000106B	40404040	4040E385		635 VXCPLINE DC C' Test # '
00001078	A7A7A7			636 VXCPNUM DC C'xxx'
0000107B	40A69996	958740C6		637 DC c' wrong FPC VXC for instruction '
0000109C	A7A7A7A7	A7A7A7A7		638 VXCPNAME DC CL8'xxxxxxxx'
000010A4	4085A797	8583A385		639 DC C' expected: VXC='
000010B5	A7A7A7			640 VXCPEXP DC C'xxx'
000010B8	6B			641 DC C','
000010B9	40998583	8589A585		642 DC C' received: VXC='
000010CA	A7A7A7			643 VXCPGOT DC C'xxx'
000010CD	4B			644 DC C'.'
		00000063	00000001	645 VXCPLNG EQU *-VXCPLINE

LOC	OBJECT CODE	ADDR1	ADDR2	STMT
				701 *****
				702 * E6TEST DSECT
				703 *****
				705 E6TEST DSECT ,
00000000	00000000			706 TSUB DC A(0) pointer to test
00000004	0000			707 TNUM DC H'00' Test Number
00000006	00			708 DC X'00'
00000007	40			709 XCSKIP DC CL1' ' Y = skip cross check
00000008	00			710 M4 DC HL1'00' m4 used
00000009	00			711 M5 DC HL1'00' m5 used
0000000A	00			712 FLG DC X'00' expected FPC flags
0000000B	00			713 VXC DC X'00' VXC expected
0000000C	00000000			714 V2ADDR DC A(0) address of v2: 16-byte packed decimal
00000010	40404040 40404040			715 OPNAME DC CL8' ' E6 name
00000018	00000000			716 RELEN DC A(0) result length
0000001C	00000000			717 READDR DC A(0) expected result address
00000020	00000000 00000000			718 DS FD gap
00000028	00000000 00000000			719 V10OUTPUT DS XL16 V1 Output
00000038	00000000 00000000			720 DS FD gap
00000040	00000000			721 FPC_R DS F FPC after instruction
00000048	00000000 00000000			722 DS FD gap
00000050	40404040 40404040			723 SKIPXC DC CL8' ' was cross check skipped?
00000058	00000000 00000000			724 DS FD gap
00000060	00000000 00000000			725 XCOUTPUT DS XL16 Cross check Output
00000070	00000000 00000000			726 DS XL16
00000080	00000000 00000000			727 DS FD gap
00000088	00000000			728 FPC_XC1 DS F 1st cross check FPC
0000008C	00000000			729 FPC_XC2 DS F 2nd cross check FPC
00000090	00000000 00000000			730 DS FD gap
00000098	00000000			731 WK1 DS F debug area
0000009C	00000000			732 WK2 DS F
000000A0				733 DS 0F
				734 **
				735 * test routine will be here (from VRR-c macro)
000011E0		00000000 00002D83		737 ZVE6TST CSECT ,
				738 DS 0F
				740 *****
				741 * Macros to help build test tables
				742 *****
				744 *
				745 * macro to generate individual test
				746 *
				747 MACRO
				748 VRR_C &INST,&M4,&M5,&FLAGS,&VXC,&SKIP
				749 .* &INST - VRR-c instruction under test
				750 .* &m4 - m4 field
				751 .* &m5 - m5 field

LOC	OBJECT CODE	ADDR1	ADDR2	STMT	
				752 .*	&flags - expected FPC flags (HEX)
				753 .*	&VXC - expected VXC (HEX)
				754 .*	&SKIP - S = skip cross check
			GBLA	&TNUM	
			SETA	&TNUM+1	
				757	
			DS	0FD	
			USING	*,R5	base for test data and test routine
				760	
			T&TNUM	DC	A(X&TNUM) address of test routine
				DC	H'&TNUM' test number
				763	
				DC	X'00'
				764	
				DC	CL1'&SKIP' Y = skip cross check
				765	
				DC	HL1'&M4' m4
				766	
				DC	HL1'&M5' m5
			FLG&TNUM	DC	X'&FLAGS' expected FPC flags
			VXC&TNUM	DC	X'&VXC' expected VXC
			V2_&TNUM	DC	A(RE&TNUM+16) address of v2: 16-byte packed decimal
				DC	CL8'&INST' instruction name
				DC	A(16) result length
				DC	A(RE&TNUM) address of expected resul
				DS	FD gap
			V10&TNUM	DS	XL16 V1 output
				DS	FD gap
			FPC_R_&TNUM	DS	F FPC after instruction
				DS	FD gap
				DC	CL8' ' was cross check skipped?
				DS	FD gap
			XCO&TNUM	DS	XL16 Cross check Output
				DS	XL16
				DS	FD gap
			FPC_XC_&TNUM	DS	F 1st cross check FPC
				DS	F 2nd cross check FPC
				DS	FD gap
				DS	F debug area
				DS	F
				788 .*	
				789 *	
			X&TNUM	DS	0F
			LFPC	FPCINIT	initialize FPC
			*	LFPC	=XL4'08000000' set trap exception
				793	
			LGF	R2,V2_&TNUM	get v2
			VLM	V22,V23,0(R2)	
				796	
			&INST	V22,V22,V23,&M4,&M5	test instruction (dest is source)
				798	
			STFPC	FPC_R_&TNUM	save FPC
			VST	V22,V10&TNUM	save instruction result
				801	
			BR	R11	return
				803	
			RE&TNUM	DS	0F expected 16 byte result
				805	
			DROP	R5	
				806	
				807	
				MEND	

LOC	OBJECT CODE	ADDR1	ADDR2	STMT
				831 *****
				832 * E6 VRR-c tests
				833 *****
				834 PRINT DATA
				835 *
				836 * E675 VCRNF - VECTOR FP CONVERT AND ROUND TO NNP
				837 *
				838 *-----
				839 * VRR-c instruction, m4, m5, flags, VXC, SKIP
				840 * NOTE: flags and VXC are HEX values
				841 * followed by
				842 * v1 - 16 byte expected result
				843 * v2 - 16 byte source
				844 * v3 - 16 byte source
				845 *-----
				846 * VCRNF - VECTOR FP CONVERT AND ROUND TO NNP
				847 *-----
				848 * Short Float -> dlfloat (with cross check dlfloat -> short float)
				849 *
				850 * some of these tests use numbers from PoP SA22-7832-13,
				851 * Figure 9-2. Examples of Floating-Point Numbers (page 9-6)
				852 *
				853 * +0 simple instruction test and 'test' test
				854 VRR_C VCRNF,0,2,00,00,N
000011E0				855+ DS 0FD
000011E0		000011E0		856+ USING *,R5 base for test data and test routine
000011E0	00001280			857+T1 DC A(X1) address of test routine
000011E4	0001			858+ DC H'1' test number
000011E6	00			859+ DC X'00'
000011E7	D5			860+ DC CL1'N' Y = skip cross check
000011E8	00			861+ DC HL1'0' m4
000011E9	02			862+ DC HL1'2' m5
000011EA	00			863+FLG1 DC X'00' expected FPC flags
000011EB	00			864+VXC1 DC X'00' expected VXC
000011EC	000012B4			865+V2_1 DC A(RE1+16) address of v2: 16-byte packed decimal
000011F0	E5C3D9D5 C6404040			866+ DC CL8'VCRNF' instruction name
000011F8	00000010			867+ DC A(16) result length
000011FC	000012A4			868+ DC A(RE1) address of expected resul
00001200	00000000 00000000			869+ DS FD gap
00001208	00000000 00000000			870+V101 DS XL16 V1 output
00001210	00000000 00000000			
00001218	00000000 00000000			871+ DS FD gap
00001220	00000000			872+FPC_R_1 DS F FPC after instruction
00001228	00000000 00000000			873+ DS FD gap
00001230	40404040 40404040			874+ DC CL8' ' was cross check skipped?
00001238	00000000 00000000			875+ DS FD gap
00001240	00000000 00000000			876+XC01 DS XL16 Cross check Output
00001248	00000000 00000000			
00001250	00000000 00000000			877+ DS XL16
00001258	00000000 00000000			
00001260	00000000 00000000			878+ DS FD gap
00001268	00000000			879+FPC_XC_1 DS F 1st cross check FPC
0000126C	00000000			880+ DS F 2nd cross check FPC
00001270	00000000 00000000			881+ DS FD gap
00001278	00000000			882+ DS F debug area
0000127C	00000000			883+ DS F

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
				884+*			
00001280				885+X1	DS	0F	
00001280	B29D 852C		0000072C	886+	LFPC	FPCINIT	initialize FPC
				887+*	LFPC	=XL4'08000000'	set trap exception
00001284	E320 500C 0014		000011EC	888+	LGF	R2,V2_1	get v2
0000128A	E767 2000 0C36		00000000	889+	VLM	V22,V23,0(R2)	
00001290	E666 7002 0E75			890+	VCRNF	V22,V22,V23,0,2	test instruction (dest is source)
00001296	B29C 9020		00001220	891+	STFPC	FPC_R_1	save FPC
0000129A	E760 9008 080E		00001208	892+	VST	V22,V101	save instruction result
000012A0	07FB			893+	BR	R11	return
000012A4				894+RE1	DS	0F	expected 16 byte result
000012A4				895+	DROP	R5	
000012A4	00000000 00000000			896	DC	XL16'0000000000000000 0000000000000000'	
000012AC	00000000 00000000						
000012B4	00000000 00000000			897	DC	XL16'0000000000000000 0000000000000000'	
000012BC	00000000 00000000						
000012C4	00000000 00000000			898	DC	XL16'0000000000000000 0000000000000000'	
000012CC	00000000 00000000						
				899			
				900 * +1, -1			
				901	VRR_C	VCRNF,0,2,00,00,N	
000012D8				902+	DS	0FD	
000012D8		000012D8		903+	USING	*,R5	base for test data and test routine
000012D8	00001378			904+T2	DC	A(X2)	address of test routine
000012DC	0002			905+	DC	H'2'	test number
000012DE	00			906+	DC	X'00'	
000012DF	D5			907+	DC	CL1'N'	Y = skip cross check
000012E0	00			908+	DC	HL1'0'	m4
000012E1	02			909+	DC	HL1'2'	m5
000012E2	00			910+FLG2	DC	X'00'	expected FPC flags
000012E3	00			911+VXC2	DC	X'00'	expected VXC
000012E4	000013AC			912+V2_2	DC	A(RE2+16)	address of v2: 16-byte packed decimal
000012E8	E5C3D9D5 C6404040			913+	DC	CL8'VCRNF'	instruction name
000012F0	00000010			914+	DC	A(16)	result length
000012F4	0000139C			915+	DC	A(RE2)	address of expected resul
000012F8	00000000 00000000			916+	DS	FD	gap
00001300	00000000 00000000			917+V102	DS	XL16	V1 output
00001308	00000000 00000000						
00001310	00000000 00000000			918+	DS	FD	gap
00001318	00000000			919+FPC_R_2	DS	F	FPC after instruction
00001320	00000000 00000000			920+	DS	FD	gap
00001328	40404040 40404040			921+	DC	CL8' '	was cross check skipped?
00001330	00000000 00000000			922+	DS	FD	gap
00001338	00000000 00000000			923+XC02	DS	XL16	Cross check Output
00001340	00000000 00000000						
00001348	00000000 00000000			924+	DS	XL16	
00001350	00000000 00000000						
00001358	00000000 00000000			925+	DS	FD	gap
00001360	00000000			926+FPC_XC_2	DS	F	1st cross check FPC
00001364	00000000			927+	DS	F	2nd cross check FPC
00001368	00000000 00000000			928+	DS	FD	gap
00001370	00000000			929+	DS	F	debug area
00001374	00000000			930+	DS	F	
				931+*			
00001378				932+X2	DS	0F	
00001378	B29D 852C		0000072C	933+	LFPC	FPCINIT	initialize FPC

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
				934+*	LFPC	=XL4'08000000'	set trap exception
0000137C	E320 500C 0014		000012E4	935+	LGF	R2,V2_2	get v2
00001382	E767 2000 0C36		00000000	936+	VLM	V22,V23,0(R2)	
00001388	E666 7002 0E75			937+	VCRNF	V22,V22,V23,0,2	test instruction (dest is source)
0000138E	B29C 5040		00001318	938+	STFPC	FPC_R_2	save FPC
00001392	E760 5028 080E		00001300	939+	VST	V22,V102	save instruction result
00001398	07FB			940+	BR	R11	return
0000139C				941+RE2	DS	0F	expected 16 byte result
0000139C				942+	DROP	R5	
0000139C	3E000000 00000000			943	DC	XL16'3E00000000000000 BE0000000000000000'	
000013A4	BE000000 00000000						
000013AC	3F800000 00000000			944	DC	XL16'3F80000000000000 0000000000000000'	
000013B4	00000000 00000000						
000013BC	BF800000 00000000			945	DC	XL16'BF80000000000000 0000000000000000'	
000013C4	00000000 00000000						
				946			
				947 * +.5, -.5			
				948	VRR_C	VCRNF,0,2,00,00,N	
000013D0				949+	DS	0FD	
000013D0		000013D0		950+	USING	*,R5	base for test data and test routine
000013D0	00001470			951+T3	DC	A(X3)	address of test routine
000013D4	0003			952+	DC	H'3'	test number
000013D6	00			953+	DC	X'00'	
000013D7	D5			954+	DC	CL1'N'	Y = skip cross check
000013D8	00			955+	DC	HL1'0'	m4
000013D9	02			956+	DC	HL1'2'	m5
000013DA	00			957+FLG3	DC	X'00'	expected FPC flags
000013DB	00			958+VXC3	DC	X'00'	expected VXC
000013DC	000014A4			959+V2_3	DC	A(RE3+16)	address of v2: 16-byte packed decimal
000013E0	E5C3D9D5 C6404040			960+	DC	CL8'VCRNF'	instruction name
000013E8	00000010			961+	DC	A(16)	result length
000013EC	00001494			962+	DC	A(RE3)	address of expected resul
000013F0	00000000 00000000			963+	DS	FD	gap
000013F8	00000000 00000000			964+V103	DS	XL16	V1 output
00001400	00000000 00000000						
00001408	00000000 00000000			965+	DS	FD	gap
00001410	00000000			966+FPC_R_3	DS	F	FPC after instruction
00001418	00000000 00000000			967+	DS	FD	gap
00001420	40404040 40404040			968+	DC	CL8' '	was cross check skipped?
00001428	00000000 00000000			969+	DS	FD	gap
00001430	00000000 00000000			970+XC03	DS	XL16	Cross check Output
00001438	00000000 00000000						
00001440	00000000 00000000			971+	DS	XL16	
00001448	00000000 00000000						
00001450	00000000 00000000			972+	DS	FD	gap
00001458	00000000			973+FPC_XC_3	DS	F	1st cross check FPC
0000145C	00000000			974+	DS	F	2nd cross check FPC
00001460	00000000 00000000			975+	DS	FD	gap
00001468	00000000			976+	DS	F	debug area
0000146C	00000000			977+	DS	F	
				978+*			
00001470				979+X3	DS	0F	
00001470	B29D 852C		0000072C	980+	LFPC	FPCINIT	initialize FPC
				981+*	LFPC	=XL4'08000000'	set trap exception
00001474	E320 500C 0014		000013DC	982+	LGF	R2,V2_3	get v2
0000147A	E767 2000 0C36		00000000	983+	VLM	V22,V23,0(R2)	

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00001480	E666 7002 0E75			984+	VCRNF	V22,V22,V23,0,2	test instruction (dest is source)
00001486	B29C 5040		00001410	985+	STFPC	FPC_R_3	save FPC
0000148A	E760 5028 080E		000013F8	986+	VST	V22,V103	save instruction result
00001490	07FB			987+	BR	R11	return
00001494				988+RE3	DS	0F	expected 16 byte result
00001494				989+	DROP	R5	
00001494	3C000000 00000000			990	DC	XL16'3C00000000000000 BC00000000000000'	
0000149C	BC000000 00000000						
000014A4	3F000000 00000000			991	DC	XL16'3F00000000000000 0000000000000000'	
000014AC	00000000 00000000						
000014B4	BF000000 00000000			992	DC	XL16'BF00000000000000 0000000000000000'	
000014BC	00000000 00000000						
				993			
				994 * +1/64, -1/64			
				995	VRR_C	VCRNF,0,2,00,00,N	
000014C8				996+	DS	0FD	
000014C8		000014C8		997+	USING	*,R5	base for test data and test routine
000014C8	00001568			998+T4	DC	A(X4)	address of test routine
000014CC	0004			999+	DC	H'4'	test number
000014CE	00			1000+	DC	X'00'	
000014CF	D5			1001+	DC	CL1'N'	Y = skip cross check
000014D0	00			1002+	DC	HL1'0'	m4
000014D1	02			1003+	DC	HL1'2'	m5
000014D2	00			1004+FLG4	DC	X'00'	expected FPC flags
000014D3	00			1005+VXC4	DC	X'00'	expected VXC
000014D4	0000159C			1006+V2_4	DC	A(RE4+16)	address of v2: 16-byte packed decimal
000014D8	E5C3D9D5 C6404040			1007+	DC	CL8'VCRNF'	instruction name
000014E0	00000010			1008+	DC	A(16)	result length
000014E4	0000158C			1009+	DC	A(RE4)	address of expected resul
000014E8	00000000 00000000			1010+	DS	FD	gap
000014F0	00000000 00000000			1011+V104	DS	XL16	V1 output
000014F8	00000000 00000000						
00001500	00000000 00000000			1012+	DS	FD	gap
00001508	00000000			1013+FPC_R_4	DS	F	FPC after instruction
00001510	00000000 00000000			1014+	DS	FD	gap
00001518	40404040 40404040			1015+	DC	CL8' '	was cross check skipped?
00001520	00000000 00000000			1016+	DS	FD	gap
00001528	00000000 00000000			1017+XC04	DS	XL16	Cross check Output
00001530	00000000 00000000						
00001538	00000000 00000000			1018+	DS	XL16	
00001540	00000000 00000000						
00001548	00000000 00000000			1019+	DS	FD	gap
00001550	00000000			1020+FPC_XC_4	DS	F	1st cross check FPC
00001554	00000000			1021+	DS	F	2nd cross check FPC
00001558	00000000 00000000			1022+	DS	FD	gap
00001560	00000000			1023+	DS	F	debug area
00001564	00000000			1024+	DS	F	
				1025+*			
00001568				1026+X4	DS	0F	
00001568	B29D 852C		0000072C	1027+	LFPC	FPCINIT	initialize FPC
				1028+*	LFPC	=XL4'08000000'	set trap exception
0000156C	E320 500C 0014		000014D4	1029+	LGF	R2,V2_4	get v2
00001572	E767 2000 0C36		00000000	1030+	VLM	V22,V23,0(R2)	
00001578	E666 7002 0E75			1031+	VCRNF	V22,V22,V23,0,2	test instruction (dest is source)
0000157E	B29C 5040		00001508	1032+	STFPC	FPC_R_4	save FPC
00001582	E760 5028 080E		000014F0	1033+	VST	V22,V104	save instruction result

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00001588	07FB			1034+	BR	R11	return
0000158C				1035+RE4	DS	0F	expected 16 byte result
0000158C				1036+	DROP	R5	
0000158C	32000000 00000000			1037	DC	XL16'	3200000000000000 B200000000000000'
00001594	B2000000 00000000						
0000159C	3C800000 00000000			1038	DC	XL16'	3C80000000000000 0000000000000000'
000015A4	00000000 00000000						
000015AC	BC800000 00000000			1039	DC	XL16'	BC80000000000000 0000000000000000'
000015B4	00000000 00000000						
				1040			
				1041 * +0, -0			
				1042	VRR_C	VCRNF, 0, 2, 00, 00, N	
000015C0				1043+	DS	0FD	
000015C0		000015C0		1044+	USING	*, R5	base for test data and test routine
000015C0	00001660			1045+T5	DC	A(X5)	address of test routine
000015C4	0005			1046+	DC	H'5'	test number
000015C6	00			1047+	DC	X'00'	
000015C7	D5			1048+	DC	CL1'N'	Y = skip cross check
000015C8	00			1049+	DC	HL1'0'	m4
000015C9	02			1050+	DC	HL1'2'	m5
000015CA	00			1051+FLG5	DC	X'00'	expected FPC flags
000015CB	00			1052+VXC5	DC	X'00'	expected VXC
000015CC	00001694			1053+V2_5	DC	A(RE5+16)	address of v2: 16-byte packed decimal
000015D0	E5C3D9D5 C6404040			1054+	DC	CL8'VCRNF'	instruction name
000015D8	00000010			1055+	DC	A(16)	result length
000015DC	00001684			1056+	DC	A(RE5)	address of expected resul
000015E0	00000000 00000000			1057+	DS	FD	gap
000015E8	00000000 00000000			1058+V105	DS	XL16	V1 output
000015F0	00000000 00000000						
000015F8	00000000 00000000			1059+	DS	FD	gap
00001600	00000000			1060+FPC_R_5	DS	F	FPC after instruction
00001608	00000000 00000000			1061+	DS	FD	gap
00001610	40404040 40404040			1062+	DC	CL8' '	was cross check skipped?
00001618	00000000 00000000			1063+	DS	FD	gap
00001620	00000000 00000000			1064+XC05	DS	XL16	Cross check Output
00001628	00000000 00000000						
00001630	00000000 00000000			1065+	DS	XL16	
00001638	00000000 00000000						
00001640	00000000 00000000			1066+	DS	FD	gap
00001648	00000000			1067+FPC_XC_5	DS	F	1st cross check FPC
0000164C	00000000			1068+	DS	F	2nd cross check FPC
00001650	00000000 00000000			1069+	DS	FD	gap
00001658	00000000			1070+	DS	F	debug area
0000165C	00000000			1071+	DS	F	
				1072+*			
00001660				1073+X5	DS	0F	
00001660	B29D 852C		0000072C	1074+	LFPC	FPCINIT	initialize FPC
				1075+*	LFPC	=XL4'08000000'	set trap exception
00001664	E320 500C 0014		000015CC	1076+	LGF	R2, V2_5	get v2
0000166A	E767 2000 0C36		00000000	1077+	VLM	V22, V23, 0(R2)	
00001670	E666 7002 0E75			1078+	VCRNF	V22, V22, V23, 0, 2	test instruction (dest is source)
00001676	B29C 5040		00001600	1079+	STFPC	FPC_R_5	save FPC
0000167A	E760 5028 080E		000015E8	1080+	VST	V22, V105	save instruction result
00001680	07FB			1081+	BR	R11	return
00001684				1082+RE5	DS	0F	expected 16 byte result
00001684				1083+	DROP	R5	

LOC	OBJECT CODE	ADDR1	ADDR2	STMT		
00001684	00000000 00000000			1084	DC	XL16'0000000000000000 8000000000000000'
0000168C	80000000 00000000					
00001694	00000000 00000000			1085	DC	XL16'0000000000000000 0000000000000000'
0000169C	00000000 00000000					
000016A4	80000000 00000000			1086	DC	XL16'8000000000000000 0000000000000000'
000016AC	00000000 00000000					
				1087		
				1088 * +15, -15		
				1089	VRR_C	VCRNF, 0, 2, 00, 00, N
000016B8				1090+	DS	0FD
000016B8		000016B8		1091+	USING	*, R5
000016B8	00001758			1092+T6	DC	A(X6)
000016BC	0006			1093+	DC	H'6'
000016BE	00			1094+	DC	X'00'
000016BF	D5			1095+	DC	CL1'N'
000016C0	00			1096+	DC	HL1'0'
000016C1	02			1097+	DC	HL1'2'
000016C2	00			1098+FLG6	DC	X'00'
000016C3	00			1099+VXC6	DC	X'00'
000016C4	0000178C			1100+V2_6	DC	A(RE6+16)
000016C8	E5C3D9D5 C6404040			1101+	DC	CL8'VCRNF'
000016D0	00000010			1102+	DC	A(16)
000016D4	0000177C			1103+	DC	A(RE6)
000016D8	00000000 00000000			1104+	DS	FD
000016E0	00000000 00000000			1105+V106	DS	XL16
000016E8	00000000 00000000					
000016F0	00000000 00000000			1106+	DS	FD
000016F8	00000000			1107+FPC_R_6	DS	F
00001700	00000000 00000000			1108+	DS	FD
00001708	40404040 40404040			1109+	DC	CL8' '
00001710	00000000 00000000			1110+	DS	FD
00001718	00000000 00000000			1111+XC06	DS	XL16
00001720	00000000 00000000					
00001728	00000000 00000000			1112+	DS	XL16
00001730	00000000 00000000					
00001738	00000000 00000000			1113+	DS	FD
00001740	00000000			1114+FPC_XC_6	DS	F
00001744	00000000			1115+	DS	F
00001748	00000000 00000000			1116+	DS	FD
00001750	00000000			1117+	DS	F
00001754	00000000			1118+	DS	F
				1119+*		
00001758				1120+X6	DS	0F
00001758	B29D 852C		0000072C	1121+	LFPC	FPCINIT
				1122+*	LFPC	=XL4'08000000'
0000175C	E320 500C 0014		000016C4	1123+	LGF	R2, V2_6
00001762	E767 2000 0C36		00000000	1124+	VLM	V22, V23, 0(R2)
00001768	E666 7002 0E75			1125+	VCRNF	V22, V22, V23, 0, 2
0000176E	B29C 5040		000016F8	1126+	STFPC	FPC_R_6
00001772	E760 5028 080E		000016E0	1127+	VST	V22, V106
00001778	07FB			1128+	BR	R11
0000177C				1129+RE6	DS	0F
0000177C				1130+	DROP	R5
0000177C	45C00000 00000000			1131	DC	XL16'45C0000000000000 C5C0000000000000'
00001784	C5C00000 00000000					
0000178C	41700000 00000000			1132	DC	XL16'4170000000000000 0000000000000000'

LOC	OBJECT CODE	ADDR1	ADDR2	STMT		
00001794	00000000 00000000					
0000179C	C1700000 00000000			1133	DC	XL16'C1700000000000000 0000000000000000'
000017A4	00000000 00000000					
				1134		
				1135	* +20/7, - 20/7	
				1136	VRR_C VCRNF,0,2,00,00,S	skip xc: lost digits
000017B0				1137+	DS	0FD
000017B0		000017B0		1138+	USING	*,R5
000017B0	00001850			1139+T7	DC	A(X7)
000017B4	0007			1140+	DC	H'7'
000017B6	00			1141+	DC	X'00'
000017B7	E2			1142+	DC	CL1'S'
000017B8	00			1143+	DC	HL1'0'
000017B9	02			1144+	DC	HL1'2'
000017BA	00			1145+FLG7	DC	X'00'
000017BB	00			1146+VXC7	DC	X'00'
000017BC	00001884			1147+V2_7	DC	A(RE7+16)
000017C0	E5C3D9D5 C6404040			1148+	DC	CL8'VCRNF'
000017C8	00000010			1149+	DC	A(16)
000017CC	00001874			1150+	DC	A(RE7)
000017D0	00000000 00000000			1151+	DS	FD
000017D8	00000000 00000000			1152+V107	DS	XL16
000017E0	00000000 00000000					
000017E8	00000000 00000000			1153+	DS	FD
000017F0	00000000			1154+FPC_R_7	DS	F
000017F8	00000000 00000000			1155+	DS	FD
00001800	40404040 40404040			1156+	DC	CL8' '
00001808	00000000 00000000			1157+	DS	FD
00001810	00000000 00000000			1158+XC07	DS	XL16
00001818	00000000 00000000					
00001820	00000000 00000000			1159+	DS	XL16
00001828	00000000 00000000					
00001830	00000000 00000000			1160+	DS	FD
00001838	00000000			1161+FPC_XC_7	DS	F
0000183C	00000000			1162+	DS	F
00001840	00000000 00000000			1163+	DS	FD
00001848	00000000			1164+	DS	F
0000184C	00000000			1165+	DS	F
				1166+*		
00001850				1167+X7	DS	0F
00001850	B29D 852C		0000072C	1168+	LFPC	FPCINIT
				1169+*	LFPC	=XL4'08000000'
00001854	E320 500C 0014		000017BC	1170+	LGF	R2,V2_7
0000185A	E767 2000 0C36		00000000	1171+	VLM	V22,V23,0(R2)
00001860	E666 7002 0E75			1172+	VCRNF	V22,V22,V23,0,2
00001866	B29C 5040		000017F0	1173+	STFPC	FPC_R_7
0000186A	E760 5028 080E		000017D8	1174+	VST	V22,V107
00001870	07FB			1175+	BR	R11
00001874				1176+RE7	DS	0F
00001874				1177+	DROP	R5
00001874	40DB0000 00000000			1178	DC	XL16'40DB000000000000 C0DB000000000000'
0000187C	C0DB0000 00000000					
00001884	4036DB6E 00000000			1179	DC	XL16'4036DB6E00000000 0000000000000000'
0000188C	00000000 00000000					
00001894	C036DB6E 00000000			1180	DC	XL16'C036DB6E00000000 0000000000000000'
0000189C	00000000 00000000					

LOC	OBJECT CODE	ADDR1	ADDR2	STMT	
				1181	
				1182 * +2 ⁽⁻¹²⁶⁾ , -2 ⁽⁻¹²⁶⁾	
				1183 VRR_C VCRNF, 0, 2, 10, 44, S	skip xc: underflow
000018A8				1184+ DS 0FD	
000018A8		000018A8		1185+ USING *, R5	base for test data and test routine
000018A8	00001948			1186+T8 DC A(X8)	address of test routine
000018AC	0008			1187+ DC H'8'	test number
000018AE	00			1188+ DC X'00'	
000018AF	E2			1189+ DC CL1'S'	Y = skip cross check
000018B0	00			1190+ DC HL1'0'	m4
000018B1	02			1191+ DC HL1'2'	m5
000018B2	10			1192+FLG8 DC X'10'	expected FPC flags
000018B3	44			1193+VXC8 DC X'44'	expected VXC
000018B4	0000197C			1194+V2_8 DC A(RE8+16)	address of v2: 16-byte packed decimal
000018B8	E5C3D9D5 C6404040			1195+ DC CL8'VCRNF'	instruction name
000018C0	00000010			1196+ DC A(16)	result length
000018C4	0000196C			1197+ DC A(RE8)	address of expected resul
000018C8	00000000 00000000			1198+ DS FD	gap
000018D0	00000000 00000000			1199+V108 DS XL16	V1 output
000018D8	00000000 00000000				
000018E0	00000000 00000000			1200+ DS FD	gap
000018E8	00000000			1201+FPC_R_8 DS F	FPC after instruction
000018F0	00000000 00000000			1202+ DS FD	gap
000018F8	40404040 40404040			1203+ DC CL8' '	was cross check skipped?
00001900	00000000 00000000			1204+ DS FD	gap
00001908	00000000 00000000			1205+XC08 DS XL16	Cross check Output
00001910	00000000 00000000				
00001918	00000000 00000000			1206+ DS XL16	
00001920	00000000 00000000				
00001928	00000000 00000000			1207+ DS FD	gap
00001930	00000000			1208+FPC_XC_8 DS F	1st cross check FPC
00001934	00000000			1209+ DS F	2nd cross check FPC
00001938	00000000 00000000			1210+ DS FD	gap
00001940	00000000			1211+ DS F	debug area
00001944	00000000			1212+ DS F	
				1213+*	
00001948				1214+X8 DS 0F	
00001948	B29D 852C		0000072C	1215+ LFPC FPCINIT	initialize FPC
				1216+* LFPC =XL4'08000000'	set trap exception
0000194C	E320 500C 0014		000018B4	1217+ LGF R2, V2_8	get v2
00001952	E767 2000 0C36		00000000	1218+ VLM V22, V23, 0(R2)	
00001958	E666 7002 0E75			1219+ VCRNF V22, V22, V23, 0, 2	test instruction (dest is source)
0000195E	B29C 5040		000018E8	1220+ STFPC FPC_R_8	save FPC
00001962	E760 5028 080E		000018D0	1221+ VST V22, V108	save instruction result
00001968	07FB			1222+ BR R11	return
0000196C				1223+RE8 DS 0F	expected 16 byte result
0000196C				1224+ DROP R5	
0000196C	00000000 00000000			1225 DC XL16'0000000000000000 8000000000000000'	
00001974	80000000 00000000				
0000197C	00800000 00000000			1226 DC XL16'0080000000000000 0000000000000000'	
00001984	00000000 00000000				
0000198C	80800000 00000000			1227 DC XL16'8080000000000000 0000000000000000'	
00001994	00000000 00000000				
				1228	
				1229 * +2 ⁽⁻¹⁴⁹⁾ , -2 ⁽⁻¹⁴⁹⁾ - subnormal	
				1230 VRR_C VCRNF, 0, 2, 10, 44, S	skip xc: underflow

LOC	OBJECT CODE	ADDR1	ADDR2	STMT		
000019A0				1231+	DS	0FD
000019A0		000019A0		1232+	USING	*,R5
000019A0	00001A40			1233+T9	DC	A(X9)
000019A4	0009			1234+	DC	H'9'
000019A6	00			1235+	DC	X'00'
000019A7	E2			1236+	DC	CL1'S'
000019A8	00			1237+	DC	HL1'0'
000019A9	02			1238+	DC	HL1'2'
000019AA	10			1239+FLG9	DC	X'10'
000019AB	44			1240+VXC9	DC	X'44'
000019AC	00001A74			1241+V2_9	DC	A(RE9+16)
000019B0	E5C3D9D5 C6404040			1242+	DC	CL8'VCRNF'
000019B8	00000010			1243+	DC	A(16)
000019BC	00001A64			1244+	DC	A(RE9)
000019C0	00000000 00000000			1245+	DS	FD
000019C8	00000000 00000000			1246+V109	DS	XL16
000019D0	00000000 00000000					
000019D8	00000000 00000000			1247+	DS	FD
000019E0	00000000			1248+FPC_R_9	DS	F
000019E8	00000000 00000000			1249+	DS	FD
000019F0	40404040 40404040			1250+	DC	CL8' '
000019F8	00000000 00000000			1251+	DS	FD
00001A00	00000000 00000000			1252+XC09	DS	XL16
00001A08	00000000 00000000					
00001A10	00000000 00000000			1253+	DS	XL16
00001A18	00000000 00000000					
00001A20	00000000 00000000			1254+	DS	FD
00001A28	00000000			1255+FPC_XC_9	DS	F
00001A2C	00000000			1256+	DS	F
00001A30	00000000 00000000			1257+	DS	FD
00001A38	00000000			1258+	DS	F
00001A3C	00000000			1259+	DS	F
				1260+*		
00001A40				1261+X9	DS	0F
00001A40	B29D 852C		0000072C	1262+	LFPC	FPCINIT
				1263+*	LFPC	=XL4'08000000'
00001A44	E320 500C 0014		000019AC	1264+	LGF	R2,V2_9
00001A4A	E767 2000 0C36		00000000	1265+	VLM	V22,V23,0(R2)
00001A50	E666 7002 0E75			1266+	VCRNF	V22,V22,V23,0,2
00001A56	B29C 5040		000019E0	1267+	STFPC	FPC_R_9
00001A5A	E760 5028 080E		000019C8	1268+	VST	V22,V109
00001A60	07FB			1269+	BR	R11
00001A64				1270+RE9	DS	0F
00001A64				1271+	DROP	R5
00001A64	00000000 00000000			1272	DC	XL16'0000000000000000 8000000000000000'
00001A6C	80000000 00000000					
00001A74	00000001 00000000			1273	DC	XL16'0000000100000000 0000000000000000'
00001A7C	00000000 00000000					
00001A84	80800001 00000000			1274	DC	XL16'8080000100000000 0000000000000000'
00001A8C	00000000 00000000					
				1275		
				1276 * +2^(128) * (1 - 2^(-24)) , - +2^(128) * (1 - 2^(-24))		
				1277	VRR_C	VCRNF,0,2,20,43,S
00001A98				1278+	DS	0FD
00001A98		00001A98		1279+	USING	*,R5
00001A98	00001B38			1280+T10	DC	A(X10)

LOC	OBJECT CODE	ADDR1	ADDR2	STMT		
00001A9C	000A			1281+	DC	H'10' test number
00001A9E	00			1282+	DC	X'00'
00001A9F	E2			1283+	DC	CL1'S' Y = skip cross check
00001AA0	00			1284+	DC	HL1'0' m4
00001AA1	02			1285+	DC	HL1'2' m5
00001AA2	20			1286+FLG10	DC	X'20' expected FPC flags
00001AA3	43			1287+VXC10	DC	X'43' expected VXC
00001AA4	00001B6C			1288+V2_10	DC	A(RE10+16) address of v2: 16-byte packed decimal
00001AA8	E5C3D9D5 C6404040			1289+	DC	CL8'VCRNF' instruction name
00001AB0	00000010			1290+	DC	A(16) result length
00001AB4	00001B5C			1291+	DC	A(RE10) address of expected resul
00001AB8	00000000 00000000			1292+	DS	FD gap
00001AC0	00000000 00000000			1293+V1010	DS	XL16 V1 output
00001AC8	00000000 00000000					
00001AD0	00000000 00000000			1294+	DS	FD gap
00001AD8	00000000			1295+FPC_R_10	DS	F FPC after instruction
00001AE0	00000000 00000000			1296+	DS	FD gap
00001AE8	40404040 40404040			1297+	DC	CL8' ' was cross check skipped?
00001AF0	00000000 00000000			1298+	DS	FD gap
00001AF8	00000000 00000000			1299+XC010	DS	XL16 Cross check Output
00001B00	00000000 00000000					
00001B08	00000000 00000000			1300+	DS	XL16
00001B10	00000000 00000000					
00001B18	00000000 00000000			1301+	DS	FD gap
00001B20	00000000			1302+FPC_XC_10	DS	F 1st cross check FPC
00001B24	00000000			1303+	DS	F 2nd cross check FPC
00001B28	00000000 00000000			1304+	DS	FD gap
00001B30	00000000			1305+	DS	F debug area
00001B34	00000000			1306+	DS	F
				1307+*		
00001B38				1308+X10	DS	0F
00001B38	B29D 852C		0000072C	1309+	LFPC	FPCINIT initialize FPC
				1310+*	LFPC	=XL4'08000000' set trap exception
00001B3C	E320 500C 0014		00001AA4	1311+	LGF	R2,V2_10 get v2
00001B42	E767 2000 0C36		00000000	1312+	VLM	V22,V23,0(R2)
00001B48	E666 7002 0E75			1313+	VCRNF	V22,V22,V23,0,2 test instruction (dest is source)
00001B4E	B29C 5040		00001AD8	1314+	STFPC	FPC_R_10 save FPC
00001B52	E760 5028 080E		00001AC0	1315+	VST	V22,V1010 save instruction result
00001B58	07FB			1316+	BR	R11 return
00001B5C				1317+RE10	DS	0F expected 16 byte result
00001B5C				1318+	DROP	R5
00001B5C	7FFE0000 00000000			1319	DC	XL16'7FFE000000000000 FFFE000000000000'
00001B64	FFFE0000 00000000					
00001B6C	7F7FFFFFFF 00000000			1320	DC	XL16'7F7FFFFFFF00000000 0000000000000000'
00001B74	00000000 00000000					
00001B7C	FF7FFFFFFF 00000000			1321	DC	XL16'FF7FFFFFFF00000000 0000000000000000'
00001B84	00000000 00000000					
				1322		
				1323 * NAN, -NAN		
				1324	VRR_C	VCRNF,0,2,80,41,S skip xc: invalid on XC
00001B90				1325+	DS	0FD
00001B90		00001B90		1326+	USING	*,R5 base for test data and test routine
00001B90	00001C30			1327+T11	DC	A(X11) address of test routine
00001B94	000B			1328+	DC	H'11' test number
00001B96	00			1329+	DC	X'00'
00001B97	E2			1330+	DC	CL1'S' Y = skip cross check

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00001B98	00			1331+	DC	HL1'0'	m4
00001B99	02			1332+	DC	HL1'2'	m5
00001B9A	80			1333+FLG11	DC	X'80'	expected FPC flags
00001B9B	41			1334+VXC11	DC	X'41'	expected VXC
00001B9C	00001C64			1335+V2_11	DC	A(RE11+16)	address of v2: 16-byte packed decimal
00001BA0	E5C3D9D5 C6404040			1336+	DC	CL8'VCRNF'	instruction name
00001BA8	00000010			1337+	DC	A(16)	result length
00001BAC	00001C54			1338+	DC	A(RE11)	address of expected resul
00001BB0	00000000 00000000			1339+	DS	FD	gap
00001BB8	00000000 00000000			1340+V1011	DS	XL16	V1 output
00001BC0	00000000 00000000						
00001BC8	00000000 00000000			1341+	DS	FD	gap
00001BD0	00000000			1342+FPC_R_11	DS	F	FPC after instruction
00001BD8	00000000 00000000			1343+	DS	FD	gap
00001BE0	40404040 40404040			1344+	DC	CL8' '	was cross check skipped?
00001BE8	00000000 00000000			1345+	DS	FD	gap
00001BF0	00000000 00000000			1346+XC011	DS	XL16	Cross check Output
00001BF8	00000000 00000000						
00001C00	00000000 00000000			1347+	DS	XL16	
00001C08	00000000 00000000						
00001C10	00000000 00000000			1348+	DS	FD	gap
00001C18	00000000			1349+FPC_XC_11	DS	F	1st cross check FPC
00001C1C	00000000			1350+	DS	F	2nd cross check FPC
00001C20	00000000 00000000			1351+	DS	FD	gap
00001C28	00000000			1352+	DS	F	debug area
00001C2C	00000000			1353+	DS	F	
				1354+*			
00001C30				1355+X11	DS	0F	
00001C30	B29D 852C		0000072C	1356+	LFPC	FPCINIT	initialize FPC
				1357+*	LFPC	=XL4'08000000'	set trap exception
00001C34	E320 500C 0014		00001B9C	1358+	LGF	R2,V2_11	get v2
00001C3A	E767 2000 0C36		00000000	1359+	VLM	V22,V23,0(R2)	
00001C40	E666 7002 0E75			1360+	VCRNF	V22,V22,V23,0,2	test instruction (dest is source)
00001C46	B29C 5040		00001BD0	1361+	STFPC	FPC_R_11	save FPC
00001C4A	E760 5028 080E		00001BB8	1362+	VST	V22,V1011	save instruction result
00001C50	07FB			1363+	BR	R11	return
00001C54				1364+RE11	DS	0F	expected 16 byte result
00001C54				1365+	DROP	R5	
00001C54	7FFF0000 00000000			1366	DC	XL16'7FFF000000000000 FFFF000000000000'	
00001C5C	FFFF0000 00000000						
00001C64	7FC00000 00000000			1367	DC	XL16'7FC0000000000000 0000000000000000'	
00001C6C	00000000 00000000						
00001C74	FFC00000 00000000			1368	DC	XL16'FFC0000000000000 0000000000000000'	
00001C7C	00000000 00000000						
				1369			
				1370 * bad m4 - inexact			
				1371	VRR_C	VCRNF,2,2,08,05,S	skip xc: inexact
00001C88				1372+	DS	0FD	
00001C88		00001C88		1373+	USING	*,R5	base for test data and test routine
00001C88	00001D28			1374+T12	DC	A(X12)	address of test routine
00001C8C	000C			1375+	DC	H'12'	test number
00001C8E	00			1376+	DC	X'00'	
00001C8F	E2			1377+	DC	CL1'S'	Y = skip cross check
00001C90	02			1378+	DC	HL1'2'	m4
00001C91	02			1379+	DC	HL1'2'	m5
00001C92	08			1380+FLG12	DC	X'08'	expected FPC flags

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00001C93	05			1381+VXC12	DC	X'05'	expected VXC
00001C94	00001D5C			1382+V2_12	DC	A(RE12+16)	address of v2: 16-byte packed decimal
00001C98	E5C3D9D5 C6404040			1383+	DC	CL8'VCRNF'	instruction name
00001CA0	00000010			1384+	DC	A(16)	result length
00001CA4	00001D4C			1385+	DC	A(RE12)	address of expected resul
00001CA8	00000000 00000000			1386+	DS	FD	gap
00001CB0	00000000 00000000			1387+V1012	DS	XL16	V1 output
00001CB8	00000000 00000000						
00001CC0	00000000 00000000			1388+	DS	FD	gap
00001CC8	00000000			1389+FPC_R_12	DS	F	FPC after instruction
00001CD0	00000000 00000000			1390+	DS	FD	gap
00001CD8	40404040 40404040			1391+	DC	CL8' '	was cross check skipped?
00001CE0	00000000 00000000			1392+	DS	FD	gap
00001CE8	00000000 00000000			1393+XC012	DS	XL16	Cross check Output
00001CF0	00000000 00000000						
00001CF8	00000000 00000000			1394+	DS	XL16	
00001D00	00000000 00000000						
00001D08	00000000 00000000			1395+	DS	FD	gap
00001D10	00000000			1396+FPC_XC_12	DS	F	1st cross check FPC
00001D14	00000000			1397+	DS	F	2nd cross check FPC
00001D18	00000000 00000000			1398+	DS	FD	gap
00001D20	00000000			1399+	DS	F	debug area
00001D24	00000000			1400+	DS	F	
				1401+*			
00001D28				1402+X12	DS	0F	
00001D28	B29D 852C		0000072C	1403+	LFPC	FPCINIT	initialize FPC
				1404+*	LFPC	=XL4'08000000'	set trap exception
00001D2C	E320 500C 0014		00001C94	1405+	LGF	R2,V2_12	get v2
00001D32	E767 2000 0C36		00000000	1406+	VLM	V22,V23,0(R2)	
00001D38	E666 7002 2E75			1407+	VCRNF	V22,V22,V23,2,2	test instruction (dest is source)
00001D3E	B29C 5040		00001CC8	1408+	STFPC	FPC_R_12	save FPC
00001D42	E760 5028 080E		00001CB0	1409+	VST	V22,V1012	save instruction result
00001D48	07FB			1410+	BR	R11	return
00001D4C				1411+RE12	DS	0F	expected 16 byte result
00001D4C				1412+	DROP	R5	
00001D4C	00000000 00000000			1413	DC	XL16'0000000000000000 0000000000000000'	
00001D54	00000000 00000000						
00001D5C	7FC00000 00000000			1414	DC	XL16'7FC0000000000000 0000000000000000'	
00001D64	00000000 00000000						
00001D6C	FFC00000 00000000			1415	DC	XL16'FFC0000000000000 0000000000000000'	
00001D74	00000000 00000000						
				1416			
				1417 * bad m5 - inexact			
				1418	VRR_C	VCRNF,0,3,08,05,S	skip xc: inexact
00001D80				1419+	DS	0FD	
00001D80		00001D80		1420+	USING	*,R5	base for test data and test routine
00001D80	00001E20			1421+T13	DC	A(X13)	address of test routine
00001D84	000D			1422+	DC	H'13'	test number
00001D86	00			1423+	DC	X'00'	
00001D87	E2			1424+	DC	CL1'S'	Y = skip cross check
00001D88	00			1425+	DC	HL1'0'	m4
00001D89	03			1426+	DC	HL1'3'	m5
00001D8A	08			1427+FLG13	DC	X'08'	expected FPC flags
00001D8B	05			1428+VXC13	DC	X'05'	expected VXC
00001D8C	00001E54			1429+V2_13	DC	A(RE13+16)	address of v2: 16-byte packed decimal
00001D90	E5C3D9D5 C6404040			1430+	DC	CL8'VCRNF'	instruction name

LOC	OBJECT CODE	ADDR1	ADDR2	STMT		
00001D98	00000010			1431+	DC	A(16) result length
00001D9C	00001E44			1432+	DC	A(RE13) address of expected resul
00001DA0	00000000 00000000			1433+	DS	FD gap
00001DA8	00000000 00000000			1434+V1013	DS	XL16 V1 output
00001DB0	00000000 00000000					
00001DB8	00000000 00000000			1435+	DS	FD gap
00001DC0	00000000			1436+FPC_R_13	DS	F FPC after instruction
00001DC8	00000000 00000000			1437+	DS	FD gap
00001DD0	40404040 40404040			1438+	DC	CL8' ' was cross check skipped?
00001DD8	00000000 00000000			1439+	DS	FD gap
00001DE0	00000000 00000000			1440+XC013	DS	XL16 Cross check Output
00001DE8	00000000 00000000					
00001DF0	00000000 00000000			1441+	DS	XL16
00001DF8	00000000 00000000					
00001E00	00000000 00000000			1442+	DS	FD gap
00001E08	00000000			1443+FPC_XC_13	DS	F 1st cross check FPC
00001E0C	00000000			1444+	DS	F 2nd cross check FPC
00001E10	00000000 00000000			1445+	DS	FD gap
00001E18	00000000			1446+	DS	F debug area
00001E1C	00000000			1447+	DS	F
				1448+*		
00001E20				1449+X13	DS	0F
00001E20	B29D 852C		0000072C	1450+	LFPC	FPCINIT initialize FPC
				1451+*	LFPC	=XL4'08000000' set trap exception
00001E24	E320 500C 0014		00001D8C	1452+	LGF	R2,V2_13 get v2
00001E2A	E767 2000 0C36		00000000	1453+	VLM	V22,V23,0(R2)
00001E30	E666 7003 0E75			1454+	VCRNF	V22,V22,V23,0,3 test instruction (dest is source)
00001E36	B29C 5040		00001DC0	1455+	STFPC	FPC_R_13 save FPC
00001E3A	E760 5028 080E		00001DA8	1456+	VST	V22,V1013 save instruction result
00001E40	07FB			1457+	BR	R11 return
00001E44				1458+RE13	DS	0F expected 16 byte result
00001E44				1459+	DROP	R5
00001E44	00000000 00000000			1460	DC	XL16'0000000000000000 0000000000000000'
00001E4C	00000000 00000000					
00001E54	7FC00000 00000000			1461	DC	XL16'7FC0000000000000 0000000000000000'
00001E5C	00000000 00000000					
00001E64	FFC00000 00000000			1462	DC	XL16'FFC0000000000000 0000000000000000'
00001E6C	00000000 00000000					
				1463		
				1464	*	Additional tests
				1465	*	-----
				1466	*	+99.1, -99.1
				1467	VRR_C	VCRNF,0,2,00,00,S skip xc: lost digits
00001E78				1468+	DS	0FD
00001E78		00001E78		1469+	USING	*,R5 base for test data and test routine
00001E78	00001F18			1470+T14	DC	A(X14) address of test routine
00001E7C	000E			1471+	DC	H'14' test number
00001E7E	00			1472+	DC	X'00'
00001E7F	E2			1473+	DC	CL1'S' Y = skip cross check
00001E80	00			1474+	DC	HL1'0' m4
00001E81	02			1475+	DC	HL1'2' m5
00001E82	00			1476+FLG14	DC	X'00' expected FPC flags
00001E83	00			1477+VXC14	DC	X'00' expected VXC
00001E84	00001F4C			1478+V2_14	DC	A(RE14+16) address of v2: 16-byte packed decimal
00001E88	E5C3D9D5 C6404040			1479+	DC	CL8'VCRNF' instruction name
00001E90	00000010			1480+	DC	A(16) result length

LOC	OBJECT CODE	ADDR1	ADDR2	STMT		
00001E94	00001F3C			1481+	DC	A(RE14)
00001E98	00000000 00000000			1482+	DS	FD
00001EA0	00000000 00000000			1483+V1014	DS	XL16
00001EA8	00000000 00000000					
00001EB0	00000000 00000000			1484+	DS	FD
00001EB8	00000000			1485+FPC_R_14	DS	F
00001EC0	00000000 00000000			1486+	DS	FD
00001EC8	40404040 40404040			1487+	DC	CL8' '
00001ED0	00000000 00000000			1488+	DS	FD
00001ED8	00000000 00000000			1489+XC014	DS	XL16
00001EE0	00000000 00000000					
00001EE8	00000000 00000000			1490+	DS	XL16
00001EF0	00000000 00000000					
00001EF8	00000000 00000000			1491+	DS	FD
00001F00	00000000			1492+FPC_XC_14	DS	F
00001F04	00000000			1493+	DS	F
00001F08	00000000 00000000			1494+	DS	FD
00001F10	00000000			1495+	DS	F
00001F14	00000000			1496+	DS	F
				1497+*		
00001F18				1498+X14	DS	0F
00001F18	B29D 852C		0000072C	1499+	LFPC	FPCINIT
				1500+*	LFPC	=XL4'08000000'
00001F1C	E320 500C 0014		00001E84	1501+	LGF	R2,V2_14
00001F22	E767 2000 0C36		00000000	1502+	VLM	V22,V23,0(R2)
00001F28	E666 7002 0E75			1503+	VCRNF	V22,V22,V23,0,2
00001F2E	B29C 5040		00001EB8	1504+	STFPC	FPC_R_14
00001F32	E760 5028 080E		00001EA0	1505+	VST	V22,V1014
00001F38	07FB			1506+	BR	R11
00001F3C				1507+RE14	DS	0F
00001F3C				1508+	DROP	R5
00001F3C	4B190000 00000000			1509	DC	XL16'4B1900000000000000 CB1900000000000000'
00001F44	CB190000 00000000					
00001F4C	42C63333 00000000			1510	DC	XL16'42C633330000000000 000000000000000000'
00001F54	00000000 00000000					
00001F5C	C2C63333 00000000			1511	DC	XL16'C2C633330000000000 000000000000000000'
00001F64	00000000 00000000					
				1512		
				1513 * +100.9, -100.9		
				1514	VRR_C	VCRNF,0,2,00,00,S
00001F70				1515+	DS	0FD
00001F70		00001F70		1516+	USING	*,R5
00001F70	00002010			1517+T15	DC	A(X15)
00001F74	000F			1518+	DC	H'15'
00001F76	00			1519+	DC	X'00'
00001F77	E2			1520+	DC	CL1'S'
00001F78	00			1521+	DC	HL1'0'
00001F79	02			1522+	DC	HL1'2'
00001F7A	00			1523+FLG15	DC	X'00'
00001F7B	00			1524+VXC15	DC	X'00'
00001F7C	00002044			1525+V2_15	DC	A(RE15+16)
00001F80	E5C3D9D5 C6404040			1526+	DC	CL8'VCRNF'
00001F88	00000010			1527+	DC	A(16)
00001F8C	00002034			1528+	DC	A(RE15)
00001F90	00000000 00000000			1529+	DS	FD
00001F98	00000000 00000000			1530+V1015	DS	XL16

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00001FA0	00000000	00000000					
00001FA8	00000000	00000000		1531+	DS	FD	gap
00001FB0	00000000			1532+FPC_R_15	DS	F	FPC after instruction
00001FB8	00000000	00000000		1533+	DS	FD	gap
00001FC0	40404040	40404040		1534+	DC	CL8' '	was cross check skipped?
00001FC8	00000000	00000000		1535+	DS	FD	gap
00001FD0	00000000	00000000		1536+XC015	DS	XL16	Cross check Output
00001FD8	00000000	00000000					
00001FE0	00000000	00000000		1537+	DS	XL16	
00001FE8	00000000	00000000					
00001FF0	00000000	00000000		1538+	DS	FD	gap
00001FF8	00000000			1539+FPC_XC_15	DS	F	1st cross check FPC
00001FFC	00000000			1540+	DS	F	2nd cross check FPC
00002000	00000000	00000000		1541+	DS	FD	gap
00002008	00000000			1542+	DS	F	debug area
0000200C	00000000			1543+	DS	F	
				1544+*			
00002010				1545+X15	DS	0F	
00002010	B29D 852C		0000072C	1546+	LFPC	FPCINIT	initialize FPC
				1547+*	LFPC	=XL4'08000000'	set trap exception
00002014	E320 500C 0014		00001F7C	1548+	LGF	R2,V2_15	get v2
0000201A	E767 2000 0C36		00000000	1549+	VLM	V22,V23,0(R2)	
00002020	E666 7002 0E75			1550+	VCRNF	V22,V22,V23,0,2	test instruction (dest is source)
00002026	B29C 5040		00001FB0	1551+	STFPC	FPC_R_15	save FPC
0000202A	E760 5028 080E		00001F98	1552+	VST	V22,V1015	save instruction result
00002030	07FB			1553+	BR	R11	return
00002034				1554+RE15	DS	0F	expected 16 byte result
00002034				1555+	DROP	R5	
00002034	4B270000	00000000		1556	DC	XL16'4B2700000000000000 CB2700000000000000'	
0000203C	CB270000	00000000					
00002044	42C9CCCD	00000000		1557	DC	XL16'42C9CCCD00000000 0000000000000000'	
0000204C	00000000	00000000					
00002054	C2C9CCCD	00000000		1558	DC	XL16'C2C9CCCD00000000 0000000000000000'	
0000205C	00000000	00000000					
				1559			
				1560 * +527.7895, -527.7895			
				1561	VRR_C	VCRNF,0,2,00,00,S	skip xc: lost digits
00002068				1562+	DS	0FD	
00002068		00002068		1563+	USING	*,R5	base for test data and test routine
00002068	00002108			1564+T16	DC	A(X16)	address of test routine
0000206C	0010			1565+	DC	H'16'	test number
0000206E	00			1566+	DC	X'00'	
0000206F	E2			1567+	DC	CL1'S'	Y = skip cross check
00002070	00			1568+	DC	HL1'0'	m4
00002071	02			1569+	DC	HL1'2'	m5
00002072	00			1570+FLG16	DC	X'00'	expected FPC flags
00002073	00			1571+VXC16	DC	X'00'	expected VXC
00002074	0000213C			1572+V2_16	DC	A(RE16+16)	address of v2: 16-byte packed decimal
00002078	E5C3D9D5 C6404040			1573+	DC	CL8'VCRNF'	instruction name
00002080	00000010			1574+	DC	A(16)	result length
00002084	0000212C			1575+	DC	A(RE16)	address of expected resul
00002088	00000000	00000000		1576+	DS	FD	gap
00002090	00000000	00000000		1577+V1016	DS	XL16	V1 output
00002098	00000000	00000000					
000020A0	00000000	00000000		1578+	DS	FD	gap
000020A8	00000000			1579+FPC_R_16	DS	F	FPC after instruction

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
000020B0	00000000 00000000			1580+	DS	FD	gap
000020B8	40404040 40404040			1581+	DC	CL8' '	was cross check skipped?
000020C0	00000000 00000000			1582+	DS	FD	gap
000020C8	00000000 00000000			1583+XC016	DS	XL16	Cross check Output
000020D0	00000000 00000000						
000020D8	00000000 00000000			1584+	DS	XL16	
000020E0	00000000 00000000						
000020E8	00000000 00000000			1585+	DS	FD	gap
000020F0	00000000			1586+FPC_XC_16	DS	F	1st cross check FPC
000020F4	00000000			1587+	DS	F	2nd cross check FPC
000020F8	00000000 00000000			1588+	DS	FD	gap
00002100	00000000			1589+	DS	F	debug area
00002104	00000000			1590+	DS	F	
				1591+*			
00002108				1592+X16	DS	0F	
00002108	B29D 852C		0000072C	1593+	LFPC	FPCINIT	initialize FPC
				1594+*	LFPC	=XL4'08000000'	set trap exception
0000210C	E320 500C 0014		00002074	1595+	LGF	R2,V2_16	get v2
00002112	E767 2000 0C36		00000000	1596+	VLM	V22,V23,0(R2)	
00002118	E666 7002 0E75			1597+	VCRNF	V22,V22,V23,0,2	test instruction (dest is source)
0000211E	B29C 5040		000020A8	1598+	STFPC	FPC_R_16	save FPC
00002122	E760 5028 080E		00002090	1599+	VST	V22,V1016	save instruction result
00002128	07FB			1600+	BR	R11	return
0000212C				1601+RE16	DS	0F	expected 16 byte result
0000212C				1602+	DROP	R5	
0000212C	50100000 00000000			1603	DC	XL16'5010000000000000 D0100000000000000'	
00002134	D0100000 00000000						
0000213C	4403F287 00000000			1604	DC	XL16'4403F28700000000 0000000000000000'	
00002144	00000000 00000000						
0000214C	C403F287 00000000			1605	DC	XL16'C403F28700000000 0000000000000000'	
00002154	00000000 00000000						
				1606			
				1607 * +1055.5790, -1055.5790			
				1608	VRR_C	VCRNF,0,2,00,00,S	skip xc: lost digits
00002160				1609+	DS	0FD	
00002160		00002160		1610+	USING	*,R5	base for test data and test routine
00002160	00002200			1611+T17	DC	A(X17)	address of test routine
00002164	0011			1612+	DC	H'17'	test number
00002166	00			1613+	DC	X'00'	
00002167	E2			1614+	DC	CL1'S'	Y = skip cross check
00002168	00			1615+	DC	HL1'0'	m4
00002169	02			1616+	DC	HL1'2'	m5
0000216A	00			1617+FLG17	DC	X'00'	expected FPC flags
0000216B	00			1618+VXC17	DC	X'00'	expected VXC
0000216C	00002234			1619+V2_17	DC	A(RE17+16)	address of v2: 16-byte packed decimal
00002170	E5C3D9D5 C6404040			1620+	DC	CL8'VCRNF'	instruction name
00002178	00000010			1621+	DC	A(16)	result length
0000217C	00002224			1622+	DC	A(RE17)	address of expected resul
00002180	00000000 00000000			1623+	DS	FD	gap
00002188	00000000 00000000			1624+V1017	DS	XL16	V1 output
00002190	00000000 00000000						
00002198	00000000 00000000			1625+	DS	FD	gap
000021A0	00000000			1626+FPC_R_17	DS	F	FPC after instruction
000021A8	00000000 00000000			1627+	DS	FD	gap
000021B0	40404040 40404040			1628+	DC	CL8' '	was cross check skipped?
000021B8	00000000 00000000			1629+	DS	FD	gap

LOC	OBJECT CODE		ADDR1	ADDR2	STMT			
000021C0	00000000	00000000			1630+XC017	DS	XL16	Cross check Output
000021C8	00000000	00000000						
000021D0	00000000	00000000			1631+	DS	XL16	
000021D8	00000000	00000000						
000021E0	00000000	00000000			1632+	DS	FD	gap
000021E8	00000000				1633+FPC_XC_17	DS	F	1st cross check FPC
000021EC	00000000				1634+	DS	F	2nd cross check FPC
000021F0	00000000	00000000			1635+	DS	FD	gap
000021F8	00000000				1636+	DS	F	debug area
000021FC	00000000				1637+	DS	F	
					1638+*			
00002200					1639+X17	DS	0F	
00002200	B29D 852C			0000072C	1640+	LFPC	FPCINIT	initialize FPC
					1641+*	LFPC	=XL4'08000000'	set trap exception
00002204	E320 500C 0014			0000216C	1642+	LGF	R2,V2_17	get v2
0000220A	E767 2000 0C36			00000000	1643+	VLM	V22,V23,0(R2)	
00002210	E666 7002 0E75				1644+	VCRNF	V22,V22,V23,0,2	test instruction (dest is source)
00002216	B29C 5040			000021A0	1645+	STFPC	FPC_R_17	save FPC
0000221A	E760 5028 080E			00002188	1646+	VST	V22,V1017	save instruction result
00002220	07FB				1647+	BR	R11	return
00002224					1648+RE17	DS	0F	expected 16 byte result
00002224					1649+	DROP	R5	
00002224	52100000 00000000				1650	DC	XL16'5210000000000000 D2100000000000000'	
0000222C	D2100000 00000000							
00002234	4483F287 00000000				1651	DC	XL16'4483F28700000000 0000000000000000'	
0000223C	00000000 00000000							
00002244	C483F287 00000000				1652	DC	XL16'C483F28700000000 0000000000000000'	
0000224C	00000000 00000000							
					1653			
					1654 * +Zero, -Zero			
					1655	VRR_C	VCRNF,0,2,00,00,N	
00002258					1656+	DS	0FD	
00002258			00002258		1657+	USING	*,R5	base for test data and test routine
00002258	000022F8				1658+T18	DC	A(X18)	address of test routine
0000225C	0012				1659+	DC	H'18'	test number
0000225E	00				1660+	DC	X'00'	
0000225F	D5				1661+	DC	CL1'N'	Y = skip cross check
00002260	00				1662+	DC	HL1'0'	m4
00002261	02				1663+	DC	HL1'2'	m5
00002262	00				1664+FLG18	DC	X'00'	expected FPC flags
00002263	00				1665+VXC18	DC	X'00'	expected VXC
00002264	0000232C				1666+V2_18	DC	A(RE18+16)	address of v2: 16-byte packed decimal
00002268	E5C3D9D5 C6404040				1667+	DC	CL8'VCRNF'	instruction name
00002270	00000010				1668+	DC	A(16)	result length
00002274	0000231C				1669+	DC	A(RE18)	address of expected resul
00002278	00000000 00000000				1670+	DS	FD	gap
00002280	00000000 00000000				1671+V1018	DS	XL16	V1 output
00002288	00000000 00000000							
00002290	00000000 00000000				1672+	DS	FD	gap
00002298	00000000				1673+FPC_R_18	DS	F	FPC after instruction
000022A0	00000000 00000000				1674+	DS	FD	gap
000022A8	40404040 40404040				1675+	DC	CL8' '	was cross check skipped?
000022B0	00000000 00000000				1676+	DS	FD	gap
000022B8	00000000 00000000				1677+XC018	DS	XL16	Cross check Output
000022C0	00000000 00000000							
000022C8	00000000 00000000				1678+	DS	XL16	

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
000022D0	00000000	00000000					
000022D8	00000000	00000000		1679+	DS	FD	gap
000022E0	00000000			1680+FPC_XC_18	DS	F	1st cross check FPC
000022E4	00000000			1681+	DS	F	2nd cross check FPC
000022E8	00000000	00000000		1682+	DS	FD	gap
000022F0	00000000			1683+	DS	F	debug area
000022F4	00000000			1684+	DS	F	
				1685+*			
000022F8				1686+X18	DS	0F	
000022F8	B29D 852C		0000072C	1687+	LFPC	FPCINIT	initialize FPC
				1688+*	LFPC	=XL4'08000000'	set trap exception
000022FC	E320 500C 0014		00002264	1689+	LGF	R2,V2_18	get v2
00002302	E767 2000 0C36		00000000	1690+	VLM	V22,V23,0(R2)	
00002308	E666 7002 0E75			1691+	VCRNF	V22,V22,V23,0,2	test instruction (dest is source)
0000230E	B29C 5040		00002298	1692+	STFPC	FPC_R_18	save FPC
00002312	E760 5028 080E		00002280	1693+	VST	V22,V1018	save instruction result
00002318	07FB			1694+	BR	R11	return
0000231C				1695+RE18	DS	0F	expected 16 byte result
0000231C				1696+	DROP	R5	
0000231C	00000000	00000000		1697	DC	XL16'0000000000000000 8000000000000000'	
00002324	80000000	00000000					
0000232C	00000000	00000000		1698	DC	XL16'0000000000000000 0000000000000000'	
00002334	00000000	00000000					
0000233C	80000000	00000000		1699	DC	XL16'8000000000000000 0000000000000000'	
00002344	00000000	00000000					
				1700			
				1701 * +Normal, -Normal			
				1702	VRR_C	VCRNF,0,2,00,00,N	skip xc: lost digits
00002350				1703+	DS	0FD	
00002350			00002350	1704+	USING	*,R5	base for test data and test routine
00002350	000023F0			1705+T19	DC	A(X19)	address of test routine
00002354	0013			1706+	DC	H'19'	test number
00002356	00			1707+	DC	X'00'	
00002357	D5			1708+	DC	CL1'N'	Y = skip cross check
00002358	00			1709+	DC	HL1'0'	m4
00002359	02			1710+	DC	HL1'2'	m5
0000235A	00			1711+FLG19	DC	X'00'	expected FPC flags
0000235B	00			1712+VXC19	DC	X'00'	expected VXC
0000235C	00002424			1713+V2_19	DC	A(RE19+16)	address of v2: 16-byte packed decimal
00002360	E5C3D9D5 C6404040			1714+	DC	CL8'VCRNF'	instruction name
00002368	00000010			1715+	DC	A(16)	result length
0000236C	00002414			1716+	DC	A(RE19)	address of expected resul
00002370	00000000	00000000		1717+	DS	FD	gap
00002378	00000000	00000000		1718+V1019	DS	XL16	V1 output
00002380	00000000	00000000					
00002388	00000000	00000000		1719+	DS	FD	gap
00002390	00000000			1720+FPC_R_19	DS	F	FPC after instruction
00002398	00000000	00000000		1721+	DS	FD	gap
000023A0	40404040 40404040			1722+	DC	CL8' '	was cross check skipped?
000023A8	00000000	00000000		1723+	DS	FD	gap
000023B0	00000000	00000000		1724+XC019	DS	XL16	Cross check Output
000023B8	00000000	00000000					
000023C0	00000000	00000000		1725+	DS	XL16	
000023C8	00000000	00000000					
000023D0	00000000	00000000		1726+	DS	FD	gap
000023D8	00000000			1727+FPC_XC_19	DS	F	1st cross check FPC

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
000023DC	00000000			1728+	DS	F	2nd cross check FPC
000023E0	00000000 00000000			1729+	DS	FD	gap
000023E8	00000000			1730+	DS	F	debug area
000023EC	00000000			1731+	DS	F	
				1732+*			
000023F0				1733+X19	DS	0F	
000023F0	B29D 852C		0000072C	1734+	LFPC	FPCINIT	initialize FPC
				1735+*	LFPC	=XL4'08000000'	set trap exception
000023F4	E320 500C 0014		0000235C	1736+	LGF	R2,V2_19	get v2
000023FA	E767 2000 0C36		00000000	1737+	VLM	V22,V23,0(R2)	
00002400	E666 7002 0E75			1738+	VCRNF	V22,V22,V23,0,2	test instruction (dest is source)
00002406	B29C 5040		00002390	1739+	STFPC	FPC_R_19	save FPC
0000240A	E760 5028 080E		00002378	1740+	VST	V22,V1019	save instruction result
00002410	07FB			1741+	BR	R11	return
00002414				1742+RE19	DS	0F	expected 16 byte result
00002414				1743+	DROP	R5	
00002414	40800000 00000000			1744	DC	XL16'4080000000000000 C080000000000000'	
0000241C	C0800000 00000000						
00002424	40200000 00000000			1745	DC	XL16'4020000000000000 0000000000000000'	
0000242C	00000000 00000000						
00002434	C0200000 00000000			1746	DC	XL16'C020000000000000 0000000000000000'	
0000243C	00000000 00000000						
				1747			
				1748 * +Subnormal, -Subnormal			
				1749	VRR_C	VCRNF,0,2,10,44,S	skip xc: lost digits
00002448				1750+	DS	0FD	
00002448		00002448		1751+	USING	*,R5	base for test data and test routine
00002448	000024E8			1752+T20	DC	A(X20)	address of test routine
0000244C	0014			1753+	DC	H'20'	test number
0000244E	00			1754+	DC	X'00'	
0000244F	E2			1755+	DC	CL1'S'	Y = skip cross check
00002450	00			1756+	DC	HL1'0'	m4
00002451	02			1757+	DC	HL1'2'	m5
00002452	10			1758+FLG20	DC	X'10'	expected FPC flags
00002453	44			1759+VXC20	DC	X'44'	expected VXC
00002454	0000251C			1760+V2_20	DC	A(RE20+16)	address of v2: 16-byte packed decimal
00002458	E5C3D9D5 C6404040			1761+	DC	CL8'VCRNF'	instruction name
00002460	00000010			1762+	DC	A(16)	result length
00002464	0000250C			1763+	DC	A(RE20)	address of expected resul
00002468	00000000 00000000			1764+	DS	FD	gap
00002470	00000000 00000000			1765+V1020	DS	XL16	V1 output
00002478	00000000 00000000						
00002480	00000000 00000000			1766+	DS	FD	gap
00002488	00000000			1767+FPC_R_20	DS	F	FPC after instruction
00002490	00000000 00000000			1768+	DS	FD	gap
00002498	40404040 40404040			1769+	DC	CL8' '	was cross check skipped?
000024A0	00000000 00000000			1770+	DS	FD	gap
000024A8	00000000 00000000			1771+XC020	DS	XL16	Cross check Output
000024B0	00000000 00000000						
000024B8	00000000 00000000			1772+	DS	XL16	
000024C0	00000000 00000000						
000024C8	00000000 00000000			1773+	DS	FD	gap
000024D0	00000000			1774+FPC_XC_20	DS	F	1st cross check FPC
000024D4	00000000			1775+	DS	F	2nd cross check FPC
000024D8	00000000 00000000			1776+	DS	FD	gap
000024E0	00000000			1777+	DS	F	debug area

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
000024E4	00000000			1778+ 1779+*	DS	F	
000024E8				1780+X20	DS	0F	
000024E8	B29D 852C		0000072C	1781+ 1782+*	LFPC	FPCINIT	initialize FPC
					LFPC	=XL4'08000000'	set trap exception
000024EC	E320 500C 0014		00002454	1783+	LGF	R2,V2_20	get v2
000024F2	E767 2000 0C36		00000000	1784+	VLM	V22,V23,0(R2)	
000024F8	E666 7002 0E75			1785+	VCRNF	V22,V22,V23,0,2	test instruction (dest is source)
000024FE	B29C 5040		00002488	1786+	STFPC	FPC_R_20	save FPC
00002502	E760 5028 080E		00002470	1787+	VST	V22,V1020	save instruction result
00002508	07FB			1788+	BR	R11	return
0000250C				1789+RE20	DS	0F	expected 16 byte result
0000250C				1790+	DROP	R5	
0000250C	00000000 00000000			1791	DC	XL16'0000000000000000 8000000000000000'	
00002514	80000000 00000000						
0000251C	007FFFFFFF 00000000			1792	DC	XL16'007FFFFFFF00000000 0000000000000000'	
00002524	00000000 00000000						
0000252C	807FFFFFFF 00000000			1793	DC	XL16'807FFFFFFF00000000 0000000000000000'	
00002534	00000000 00000000						
				1794			
				1795 * +Infinity, -Infinity			
				1796	VRR_C	VCRNF,0,2,20,43,S	skip xc: lost digits
00002540				1797+	DS	0FD	
00002540		00002540		1798+	USING	*,R5	base for test data and test routine
00002540	000025E0			1799+T21	DC	A(X21)	address of test routine
00002544	0015			1800+	DC	H'21'	test number
00002546	00			1801+	DC	X'00'	
00002547	E2			1802+	DC	CL1'S'	Y = skip cross check
00002548	00			1803+	DC	HL1'0'	m4
00002549	02			1804+	DC	HL1'2'	m5
0000254A	20			1805+FLG21	DC	X'20'	expected FPC flags
0000254B	43			1806+VXC21	DC	X'43'	expected VXC
0000254C	00002614			1807+V2_21	DC	A(RE21+16)	address of v2: 16-byte packed decimal
00002550	E5C3D9D5 C6404040			1808+	DC	CL8'VCRNF'	instruction name
00002558	00000010			1809+	DC	A(16)	result length
0000255C	00002604			1810+	DC	A(RE21)	address of expected resul
00002560	00000000 00000000			1811+	DS	FD	gap
00002568	00000000 00000000			1812+V1021	DS	XL16	V1 output
00002570	00000000 00000000						
00002578	00000000 00000000			1813+	DS	FD	gap
00002580	00000000			1814+FPC_R_21	DS	F	FPC after instruction
00002588	00000000 00000000			1815+	DS	FD	gap
00002590	40404040 40404040			1816+	DC	CL8' '	was cross check skipped?
00002598	00000000 00000000			1817+	DS	FD	gap
000025A0	00000000 00000000			1818+XC021	DS	XL16	Cross check Output
000025A8	00000000 00000000						
000025B0	00000000 00000000			1819+	DS	XL16	
000025B8	00000000 00000000						
000025C0	00000000 00000000			1820+	DS	FD	gap
000025C8	00000000			1821+FPC_XC_21	DS	F	1st cross check FPC
000025CC	00000000			1822+	DS	F	2nd cross check FPC
000025D0	00000000 00000000			1823+	DS	FD	gap
000025D8	00000000			1824+	DS	F	debug area
000025DC	00000000			1825+	DS	F	
				1826+*			
000025E0				1827+X21	DS	0F	

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
000025E0	B29D 852C		0000072C	1828+	LFPC	FPCINIT	initialize FPC
				1829+*	LFPC	=XL4'08000000'	set trap exception
000025E4	E320 500C 0014		0000254C	1830+	LGF	R2,V2_21	get v2
000025EA	E767 2000 0C36		00000000	1831+	VLM	V22,V23,0(R2)	
000025F0	E666 7002 0E75			1832+	VCRNF	V22,V22,V23,0,2	test instruction (dest is source)
000025F6	B29C 5040		00002580	1833+	STFPC	FPC_R_21	save FPC
000025FA	E760 5028 080E		00002568	1834+	VST	V22,V1021	save instruction result
00002600	07FB			1835+	BR	R11	return
00002604				1836+RE21	DS	0F	expected 16 byte result
00002604				1837+	DROP	R5	
00002604	7FFF0000 00000000			1838	DC	XL16'7FFF000000000000 FFFF000000000000'	
0000260C	FFFF0000 00000000						
00002614	7F800000 00000000			1839	DC	XL16'7F80000000000000 0000000000000000'	
0000261C	00000000 00000000						
00002624	FF800000 00000000			1840	DC	XL16'FF80000000000000 0000000000000000'	
0000262C	00000000 00000000						
				1841			
				1842 * +QNaN, -QNaN			
				1843	VRR_C	VCRNF,0,2,80,41,S	skip xc: lost digits
00002638				1844+	DS	0FD	
00002638		00002638		1845+	USING	*,R5	base for test data and test routine
00002638	000026D8			1846+T22	DC	A(X22)	address of test routine
0000263C	0016			1847+	DC	H'22'	test number
0000263E	00			1848+	DC	X'00'	
0000263F	E2			1849+	DC	CL1'S'	Y = skip cross check
00002640	00			1850+	DC	HL1'0'	m4
00002641	02			1851+	DC	HL1'2'	m5
00002642	80			1852+FLG22	DC	X'80'	expected FPC flags
00002643	41			1853+VXC22	DC	X'41'	expected VXC
00002644	0000270C			1854+V2_22	DC	A(RE22+16)	address of v2: 16-byte packed decimal
00002648	E5C3D9D5 C6404040			1855+	DC	CL8'VCRNF'	instruction name
00002650	00000010			1856+	DC	A(16)	result length
00002654	000026FC			1857+	DC	A(RE22)	address of expected resul
00002658	00000000 00000000			1858+	DS	FD	gap
00002660	00000000 00000000			1859+V1022	DS	XL16	V1 output
00002668	00000000 00000000						
00002670	00000000 00000000			1860+	DS	FD	gap
00002678	00000000			1861+FPC_R_22	DS	F	FPC after instruction
00002680	00000000 00000000			1862+	DS	FD	gap
00002688	40404040 40404040			1863+	DC	CL8' '	was cross check skipped?
00002690	00000000 00000000			1864+	DS	FD	gap
00002698	00000000 00000000			1865+XC022	DS	XL16	Cross check Output
000026A0	00000000 00000000						
000026A8	00000000 00000000			1866+	DS	XL16	
000026B0	00000000 00000000						
000026B8	00000000 00000000			1867+	DS	FD	gap
000026C0	00000000			1868+FPC_XC_22	DS	F	1st cross check FPC
000026C4	00000000			1869+	DS	F	2nd cross check FPC
000026C8	00000000 00000000			1870+	DS	FD	gap
000026D0	00000000			1871+	DS	F	debug area
000026D4	00000000			1872+	DS	F	
				1873+*			
000026D8				1874+X22	DS	0F	
000026D8	B29D 852C		0000072C	1875+	LFPC	FPCINIT	initialize FPC
				1876+*	LFPC	=XL4'08000000'	set trap exception
000026DC	E320 500C 0014		00002644	1877+	LGF	R2,V2_22	get v2

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
000026E2	E767 2000 0C36		00000000	1878+	VLM	V22,V23,0(R2)	
000026E8	E666 7002 0E75			1879+	VCRNF	V22,V22,V23,0,2	test instruction (dest is source)
000026EE	B29C 5040		00002678	1880+	STFPC	FPC_R_22	save FPC
000026F2	E760 5028 080E		00002660	1881+	VST	V22,V1022	save instruction result
000026F8	07FB			1882+	BR	R11	return
000026FC				1883+RE22	DS	0F	expected 16 byte result
000026FC				1884+	DROP	R5	
000026FC	7FFF0000 00000000			1885	DC	XL16'7FFF000000000000 FFFF000000000000'	
00002704	FFFF0000 00000000						
0000270C	7FC10000 00000000			1886	DC	XL16'7FC1000000000000 0000000000000000'	
00002714	00000000 00000000						
0000271C	FFC10000 00000000			1887	DC	XL16'FFC1000000000000 0000000000000000'	
00002724	00000000 00000000						
				1888			
				1889 * +SNaN, -SNaN			
				1890	VRR_C	VCRNF,0,2,80,41,S	skip xc: lost digits
00002730				1891+	DS	0FD	
00002730		00002730		1892+	USING	*,R5	base for test data and test routine
00002730	000027D0			1893+T23	DC	A(X23)	address of test routine
00002734	0017			1894+	DC	H'23'	test number
00002736	00			1895+	DC	X'00'	
00002737	E2			1896+	DC	CL1'S'	Y = skip cross check
00002738	00			1897+	DC	HL1'0'	m4
00002739	02			1898+	DC	HL1'2'	m5
0000273A	80			1899+FLG23	DC	X'80'	expected FPC flags
0000273B	41			1900+VXC23	DC	X'41'	expected VXC
0000273C	00002804			1901+V2_23	DC	A(RE23+16)	address of v2: 16-byte packed decimal
00002740	E5C3D9D5 C6404040			1902+	DC	CL8'VCRNF'	instruction name
00002748	00000010			1903+	DC	A(16)	result length
0000274C	000027F4			1904+	DC	A(RE23)	address of expected resul
00002750	00000000 00000000			1905+	DS	FD	gap
00002758	00000000 00000000			1906+V1023	DS	XL16	V1 output
00002760	00000000 00000000						
00002768	00000000 00000000			1907+	DS	FD	gap
00002770	00000000			1908+FPC_R_23	DS	F	FPC after instruction
00002778	00000000 00000000			1909+	DS	FD	gap
00002780	40404040 40404040			1910+	DC	CL8' '	was cross check skipped?
00002788	00000000 00000000			1911+	DS	FD	gap
00002790	00000000 00000000			1912+XC023	DS	XL16	Cross check Output
00002798	00000000 00000000						
000027A0	00000000 00000000			1913+	DS	XL16	
000027A8	00000000 00000000						
000027B0	00000000 00000000			1914+	DS	FD	gap
000027B8	00000000			1915+FPC_XC_23	DS	F	1st cross check FPC
000027BC	00000000			1916+	DS	F	2nd cross check FPC
000027C0	00000000 00000000			1917+	DS	FD	gap
000027C8	00000000			1918+	DS	F	debug area
000027CC	00000000			1919+	DS	F	
				1920+*			
000027D0				1921+X23	DS	0F	
000027D0	B29D 852C		0000072C	1922+	LFPC	FPCINIT	initialize FPC
				1923+*	LFPC	=XL4'08000000'	set trap exception
000027D4	E320 500C 0014		0000273C	1924+	LGF	R2,V2_23	get v2
000027DA	E767 2000 0C36		00000000	1925+	VLM	V22,V23,0(R2)	
000027E0	E666 7002 0E75			1926+	VCRNF	V22,V22,V23,0,2	test instruction (dest is source)
000027E6	B29C 5040		00002770	1927+	STFPC	FPC_R_23	save FPC

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
000027EA	E760 5028 080E		00002758	1928+	VST	V22,V1023	save instruction result
000027F0	07FB			1929+	BR	R11	return
000027F4				1930+RE23	DS	0F	expected 16 byte result
000027F4				1931+	DROP	R5	
000027F4	7FFF0000 00000000			1932	DC	XL16'7FFF000000000000 FFFF000000000000'	
000027FC	FFFF0000 00000000						
00002804	7F810000 00000000			1933	DC	XL16'7F81000000000000 0000000000000000'	
0000280C	00000000 00000000						
00002814	FF810000 00000000			1934	DC	XL16'FF81000000000000 0000000000000000'	
0000281C	00000000 00000000						
				1935			
				1936	* Additional tests - multiple exceptions		
				1937	*-----		
				1938			
				1939	* +Subnormal, -Infinity		
				1940	VRR_C	VCRNF,0,2,30,43,S	skip xc: lost digits
00002828				1941+	DS	0FD	
00002828		00002828		1942+	USING	*,R5	base for test data and test routine
00002828	000028C8			1943+T24	DC	A(X24)	address of test routine
0000282C	0018			1944+	DC	H'24'	test number
0000282E	00			1945+	DC	X'00'	
0000282F	E2			1946+	DC	CL1'S'	Y = skip cross check
00002830	00			1947+	DC	HL1'0'	m4
00002831	02			1948+	DC	HL1'2'	m5
00002832	30			1949+FLG24	DC	X'30'	expected FPC flags
00002833	43			1950+VXC24	DC	X'43'	expected VXC
00002834	000028FC			1951+V2_24	DC	A(RE24+16)	address of v2: 16-byte packed decimal
00002838	E5C3D9D5 C6404040			1952+	DC	CL8'VCRNF'	instruction name
00002840	00000010			1953+	DC	A(16)	result length
00002844	000028EC			1954+	DC	A(RE24)	address of expected resul
00002848	00000000 00000000			1955+	DS	FD	gap
00002850	00000000 00000000			1956+V1024	DS	XL16	V1 output
00002858	00000000 00000000						
00002860	00000000 00000000			1957+	DS	FD	gap
00002868	00000000			1958+FPC_R_24	DS	F	FPC after instruction
00002870	00000000 00000000			1959+	DS	FD	gap
00002878	40404040 40404040			1960+	DC	CL8' '	was cross check skipped?
00002880	00000000 00000000			1961+	DS	FD	gap
00002888	00000000 00000000			1962+XC024	DS	XL16	Cross check Output
00002890	00000000 00000000						
00002898	00000000 00000000			1963+	DS	XL16	
000028A0	00000000 00000000						
000028A8	00000000 00000000			1964+	DS	FD	gap
000028B0	00000000			1965+FPC_XC_24	DS	F	1st cross check FPC
000028B4	00000000			1966+	DS	F	2nd cross check FPC
000028B8	00000000 00000000			1967+	DS	FD	gap
000028C0	00000000			1968+	DS	F	debug area
000028C4	00000000			1969+	DS	F	
				1970+*			
000028C8				1971+X24	DS	0F	
000028C8	B29D 852C		0000072C	1972+	LFPC	FPCINIT	initialize FPC
				1973+*	LFPC	=XL4'08000000'	set trap exception
000028CC	E320 500C 0014		00002834	1974+	LGF	R2,V2_24	get v2
000028D2	E767 2000 0C36		00000000	1975+	VLM	V22,V23,0(R2)	
000028D8	E666 7002 0E75			1976+	VCRNF	V22,V22,V23,0,2	test instruction (dest is source)
000028DE	B29C 5040		00002868	1977+	STFPC	FPC_R_24	save FPC

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
000028E2	E760 5028 080E		00002850	1978+	VST	V22,V1024	save instruction result
000028E8	07FB			1979+	BR	R11	return
000028EC				1980+RE24	DS	0F	expected 16 byte result
000028EC				1981+	DROP	R5	
000028EC	00000000 00000000			1982	DC	XL16'0000000000000000 FFFF000000000000'	
000028F4	FFFF0000 00000000						
000028FC	007FFFFFFF 00000000		1983		DC	XL16'007FFFFFFF00000000 0000000000000000'	
00002904	00000000 00000000						
0000290C	FF800000 00000000		1984		DC	XL16'FF80000000000000 0000000000000000'	
00002914	00000000 00000000						
				1985			
				1986 *	-Infinity, +Subnormal		
				1987	VRR_C	VCRNF,0,2,30,64,S	skip xc: lost digits
00002920				1988+	DS	0FD	
00002920		00002920		1989+	USING	*,R5	base for test data and test routine
00002920	000029C0			1990+T25	DC	A(X25)	address of test routine
00002924	0019			1991+	DC	H'25'	test number
00002926	00			1992+	DC	X'00'	
00002927	E2			1993+	DC	CL1'S'	Y = skip cross check
00002928	00			1994+	DC	HL1'0'	m4
00002929	02			1995+	DC	HL1'2'	m5
0000292A	30			1996+FLG25	DC	X'30'	expected FPC flags
0000292B	64			1997+VXC25	DC	X'64'	expected VXC
0000292C	000029F4			1998+V2_25	DC	A(RE25+16)	address of v2: 16-byte packed decimal
00002930	E5C3D9D5 C6404040			1999+	DC	CL8'VCRNF'	instruction name
00002938	00000010			2000+	DC	A(16)	result length
0000293C	000029E4			2001+	DC	A(RE25)	address of expected resul
00002940	00000000 00000000			2002+	DS	FD	gap
00002948	00000000 00000000			2003+V1025	DS	XL16	V1 output
00002950	00000000 00000000						
00002958	00000000 00000000			2004+	DS	FD	gap
00002960	00000000			2005+FPC_R_25	DS	F	FPC after instruction
00002968	00000000 00000000			2006+	DS	FD	gap
00002970	40404040 40404040			2007+	DC	CL8' '	was cross check skipped?
00002978	00000000 00000000			2008+	DS	FD	gap
00002980	00000000 00000000			2009+XC025	DS	XL16	Cross check Output
00002988	00000000 00000000						
00002990	00000000 00000000			2010+	DS	XL16	
00002998	00000000 00000000						
000029A0	00000000 00000000			2011+	DS	FD	gap
000029A8	00000000			2012+FPC_XC_25	DS	F	1st cross check FPC
000029AC	00000000			2013+	DS	F	2nd cross check FPC
000029B0	00000000 00000000			2014+	DS	FD	gap
000029B8	00000000			2015+	DS	F	debug area
000029BC	00000000			2016+	DS	F	
				2017+*			
000029C0				2018+X25	DS	0F	
000029C0	B29D 852C		0000072C	2019+	LFPC	FPCINIT	initialize FPC
				2020+*	LFPC	=XL4'08000000'	set trap exception
000029C4	E320 500C 0014		0000292C	2021+	LGF	R2,V2_25	get v2
000029CA	E767 2000 0C36		00000000	2022+	VLM	V22,V23,0(R2)	
000029D0	E666 7002 0E75			2023+	VCRNF	V22,V22,V23,0,2	test instruction (dest is source)
000029D6	B29C 5040		00002960	2024+	STFPC	FPC_R_25	save FPC
000029DA	E760 5028 080E		00002948	2025+	VST	V22,V1025	save instruction result
000029E0	07FB			2026+	BR	R11	return
000029E4				2027+RE25	DS	0F	expected 16 byte result

LOC	OBJECT CODE	ADDR1	ADDR2	STMT		
000029E4				2028+	DROP R5	
000029E4	00000000 00000000			2029	DC	XL16'0000000000000000 FFFF000000000000'
000029EC	FFFF0000 00000000					
000029F4	00000000 00000000			2030	DC	XL16'0000000000000000 0000000000000000'
000029FC	00000000 00000000					
00002A04	FF800000 00000000			2031	DC	XL16'FF80000000000000 007FFFFFFF00000000'
00002A0C	007FFFFFFF 00000000					
				2032		
				2033	* +Infinity, -QNaN	
				2034	VRR_C	VCRNF,0,2,A0,41,S skip xc: lost digits
00002A18				2035+	DS	0FD
00002A18		00002A18		2036+	USING	*,R5 base for test data and test routine
00002A18	00002AB8			2037+T26	DC	A(X26) address of test routine
00002A1C	001A			2038+	DC	H'26' test number
00002A1E	00			2039+	DC	X'00'
00002A1F	E2			2040+	DC	CL1'S' Y = skip cross check
00002A20	00			2041+	DC	HL1'0' m4
00002A21	02			2042+	DC	HL1'2' m5
00002A22	A0			2043+FLG26	DC	X'A0' expected FPC flags
00002A23	41			2044+VXC26	DC	X'41' expected VXC
00002A24	00002AEC			2045+V2_26	DC	A(RE26+16) address of v2: 16-byte packed decimal
00002A28	E5C3D9D5 C6404040			2046+	DC	CL8'VCRNF' instruction name
00002A30	00000010			2047+	DC	A(16) result length
00002A34	00002ADC			2048+	DC	A(RE26) address of expected resul
00002A38	00000000 00000000			2049+	DS	FD gap
00002A40	00000000 00000000			2050+V1026	DS	XL16 V1 output
00002A48	00000000 00000000					
00002A50	00000000 00000000			2051+	DS	FD gap
00002A58	00000000			2052+FPC_R_26	DS	F FPC after instruction
00002A60	00000000 00000000			2053+	DS	FD gap
00002A68	40404040 40404040			2054+	DC	CL8' ' was cross check skipped?
00002A70	00000000 00000000			2055+	DS	FD gap
00002A78	00000000 00000000			2056+XC026	DS	XL16 Cross check Output
00002A80	00000000 00000000					
00002A88	00000000 00000000			2057+	DS	XL16
00002A90	00000000 00000000					
00002A98	00000000 00000000			2058+	DS	FD gap
00002AA0	00000000			2059+FPC_XC_26	DS	F 1st cross check FPC
00002AA4	00000000			2060+	DS	F 2nd cross check FPC
00002AA8	00000000 00000000			2061+	DS	FD gap
00002AB0	00000000			2062+	DS	F debug area
00002AB4	00000000			2063+	DS	F
				2064+*		
00002AB8				2065+X26	DS	0F
00002AB8	B29D 852C		0000072C	2066+	LFPC	FPCINIT initialize FPC
				2067+*	LFPC	=XL4'08000000' set trap exception
00002ABC	E320 500C 0014		00002A24	2068+	LGF	R2,V2_26 get v2
00002AC2	E767 2000 0C36		00000000	2069+	VLM	V22,V23,0(R2)
00002AC8	E666 7002 0E75			2070+	VCRNF	V22,V22,V23,0,2 test instruction (dest is source)
00002ACE	B29C 5040		00002A58	2071+	STFPC	FPC_R_26 save FPC
00002AD2	E760 5028 080E		00002A40	2072+	VST	V22,V1026 save instruction result
00002AD8	07FB			2073+	BR	R11 return
00002ADC				2074+RE26	DS	0F expected 16 byte result
00002ADC	7FFF0000 00000000			2075+	DROP	R5
00002AE4	FFFF0000 00000000			2076	DC	XL16'7FFF000000000000 FFFF000000000000'

LOC	OBJECT CODE	ADDR1	ADDR2	STMT		
00002AEC	7F800000 00000000			2077	DC	XL16'7F80000000000000 0000000000000000'
00002AF4	00000000 00000000					
00002AFC	FFC10000 00000000			2078	DC	XL16'FFC1000000000000 0000000000000000'
00002B04	00000000 00000000					
				2079		
				2080	* +Infinity, -QNaN, +Infinity	
				2081	VRR_C	VCRNF,0,2,A0,63,S skip xc: lost digits
00002B10				2082+	DS	0FD
00002B10		00002B10		2083+	USING	*,R5 base for test data and test routine
00002B10	00002BB0			2084+T27	DC	A(X27) address of test routine
00002B14	001B			2085+	DC	H'27' test number
00002B16	00			2086+	DC	X'00'
00002B17	E2			2087+	DC	CL1'S' Y = skip cross check
00002B18	00			2088+	DC	HL1'0' m4
00002B19	02			2089+	DC	HL1'2' m5
00002B1A	A0			2090+FLG27	DC	X'A0' expected FPC flags
00002B1B	63			2091+VXC27	DC	X'63' expected VXC
00002B1C	00002BE4			2092+V2_27	DC	A(RE27+16) address of v2: 16-byte packed decimal
00002B20	E5C3D9D5 C6404040			2093+	DC	CL8'VCRNF' instruction name
00002B28	00000010			2094+	DC	A(16) result length
00002B2C	00002BD4			2095+	DC	A(RE27) address of expected resul
00002B30	00000000 00000000			2096+	DS	FD gap
00002B38	00000000 00000000			2097+V1027	DS	XL16 V1 output
00002B40	00000000 00000000					
00002B48	00000000 00000000			2098+	DS	FD gap
00002B50	00000000			2099+FPC_R_27	DS	F FPC after instruction
00002B58	00000000 00000000			2100+	DS	FD gap
00002B60	40404040 40404040			2101+	DC	CL8' ' was cross check skipped?
00002B68	00000000 00000000			2102+	DS	FD gap
00002B70	00000000 00000000			2103+XC027	DS	XL16 Cross check Output
00002B78	00000000 00000000					
00002B80	00000000 00000000			2104+	DS	XL16
00002B88	00000000 00000000					
00002B90	00000000 00000000			2105+	DS	FD gap
00002B98	00000000			2106+FPC_XC_27	DS	F 1st cross check FPC
00002B9C	00000000			2107+	DS	F 2nd cross check FPC
00002BA0	00000000 00000000			2108+	DS	FD gap
00002BA8	00000000			2109+	DS	F debug area
00002BAC	00000000			2110+	DS	F
				2111+*		
00002BB0				2112+X27	DS	0F
00002BB0	B29D 852C		0000072C	2113+	LFPC	FPCINIT initialize FPC
				2114+*	LFPC	=XL4'08000000' set trap exception
00002BB4	E320 500C 0014		00002B1C	2115+	LGF	R2,V2_27 get v2
00002BBA	E767 2000 0C36		00000000	2116+	VLM	V22,V23,0(R2)
00002BC0	E666 7002 0E75			2117+	VCRNF	V22,V22,V23,0,2 test instruction (dest is source)
00002BC6	B29C 5040		00002B50	2118+	STFPC	FPC_R_27 save FPC
00002BCA	E760 5028 080E		00002B38	2119+	VST	V22,V1027 save instruction result
00002BD0	07FB			2120+	BR	R11 return
00002BD4				2121+RE27	DS	0F expected 16 byte result
00002BD4				2122+	DROP	R5
00002BD4	7FFF0000 00000000			2123	DC	XL16'7FFF000000000000 FFFF00007FFF0000'
00002BDC	FFFF0000 7FFF0000					
00002BE4	7F800000 00000000			2124	DC	XL16'7F80000000000000 0000000000000000'
00002BEC	00000000 00000000					
00002BF4	FFC10000 00000000			2125	DC	XL16'FFC1000000000000 7F80000000000000'

LOC	OBJECT CODE	ADDR1	ADDR2	STMT	
00002BFC	7F800000 00000000			2126	
				2127 * +Infinity, -QNaN, +Subnormal	
				2128 VRR_C VCRNF,0,2,B0,64,S	skip xc: lost digits
00002C08				2129+ DS 0FD	
00002C08		00002C08		2130+ USING *,R5	base for test data and test routine
00002C08	00002CA8			2131+T28 DC A(X28)	address of test routine
00002C0C	001C			2132+ DC H'28'	test number
00002C0E	00			2133+ DC X'00'	
00002C0F	E2			2134+ DC CL1'S'	Y = skip cross check
00002C10	00			2135+ DC HL1'0'	m4
00002C11	02			2136+ DC HL1'2'	m5
00002C12	B0			2137+FLG28 DC X'B0'	expected FPC flags
00002C13	64			2138+VXC28 DC X'64'	expected VXC
00002C14	00002CDC			2139+V2_28 DC A(RE28+16)	address of v2: 16-byte packed decimal
00002C18	E5C3D9D5 C6404040			2140+ DC CL8'VCRNF'	instruction name
00002C20	00000010			2141+ DC A(16)	result length
00002C24	00002CCC			2142+ DC A(RE28)	address of expected resul
00002C28	00000000 00000000			2143+ DS FD	gap
00002C30	00000000 00000000			2144+V1028 DS XL16	V1 output
00002C38	00000000 00000000				
00002C40	00000000 00000000			2145+ DS FD	gap
00002C48	00000000			2146+FPC_R_28 DS F	FPC after instruction
00002C50	00000000 00000000			2147+ DS FD	gap
00002C58	40404040 40404040			2148+ DC CL8' '	was cross check skipped?
00002C60	00000000 00000000			2149+ DS FD	gap
00002C68	00000000 00000000			2150+XC028 DS XL16	Cross check Output
00002C70	00000000 00000000				
00002C78	00000000 00000000			2151+ DS XL16	
00002C80	00000000 00000000				
00002C88	00000000 00000000			2152+ DS FD	gap
00002C90	00000000			2153+FPC_XC_28 DS F	1st cross check FPC
00002C94	00000000			2154+ DS F	2nd cross check FPC
00002C98	00000000 00000000			2155+ DS FD	gap
00002CA0	00000000			2156+ DS F	debug area
00002CA4	00000000			2157+ DS F	
				2158+*	
00002CA8				2159+X28 DS 0F	
00002CA8	B29D 852C		0000072C	2160+ LFPC FPCINIT	initialize FPC
				2161+* LFPC =XL4'08000000'	set trap exception
00002CAC	E320 500C 0014		00002C14	2162+ LGF R2,V2_28	get v2
00002CB2	E767 2000 0C36		00000000	2163+ VLM V22,V23,0(R2)	
00002CB8	E666 7002 0E75			2164+ VCRNF V22,V22,V23,0,2	test instruction (dest is source)
00002CBE	B29C 5040		00002C48	2165+ STFPC FPC_R_28	save FPC
00002CC2	E760 5028 080E		00002C30	2166+ VST V22,V1028	save instruction result
00002CC8	07FB			2167+ BR R11	return
00002CCC				2168+RE28 DS 0F	expected 16 byte result
00002CCC				2169+ DROP R5	
00002CCC	7FFF0000 00000000			2170 DC XL16'7FFF000000000000 FFFF000000000000'	
00002CD4	FFFF0000 00000000				
00002CDC	7F800000 00000000			2171 DC XL16'7F80000000000000 0000000000000000'	
00002CE4	00000000 00000000				
00002CEC	FFC10000 00000000			2172 DC XL16'FFC1000000000000 007FFFFFFF00000000'	
00002CF4	007FFFFFFF 00000000				
				2173	
				2174	

LOC	OBJECT CODE	ADDR1	ADDR2	STMT				
00002CFC	00000000			2175	DC	F'0'	END OF TABLE	
00002D00	00000000			2176	DC	F'0'		
				2177 *				
				2178 * table of pointers to individual tests				
				2179 *				
00002D04				2180 E6TESTS	DS	0F		
				2181	PTTABLE			
00002D04				2182+TTABLE	DS	0F		
00002D04	000011E0			2183+	DC	A(T1)	TEST &CUR	
00002D08	000012D8			2184+	DC	A(T2)	TEST &CUR	
00002D0C	000013D0			2185+	DC	A(T3)	TEST &CUR	
00002D10	000014C8			2186+	DC	A(T4)	TEST &CUR	
00002D14	000015C0			2187+	DC	A(T5)	TEST &CUR	
00002D18	000016B8			2188+	DC	A(T6)	TEST &CUR	
00002D1C	000017B0			2189+	DC	A(T7)	TEST &CUR	
00002D20	000018A8			2190+	DC	A(T8)	TEST &CUR	
00002D24	000019A0			2191+	DC	A(T9)	TEST &CUR	
00002D28	00001A98			2192+	DC	A(T10)	TEST &CUR	
00002D2C	00001B90			2193+	DC	A(T11)	TEST &CUR	
00002D30	00001C88			2194+	DC	A(T12)	TEST &CUR	
00002D34	00001D80			2195+	DC	A(T13)	TEST &CUR	
00002D38	00001E78			2196+	DC	A(T14)	TEST &CUR	
00002D3C	00001F70			2197+	DC	A(T15)	TEST &CUR	
00002D40	00002068			2198+	DC	A(T16)	TEST &CUR	
00002D44	00002160			2199+	DC	A(T17)	TEST &CUR	
00002D48	00002258			2200+	DC	A(T18)	TEST &CUR	
00002D4C	00002350			2201+	DC	A(T19)	TEST &CUR	
00002D50	00002448			2202+	DC	A(T20)	TEST &CUR	
00002D54	00002540			2203+	DC	A(T21)	TEST &CUR	
00002D58	00002638			2204+	DC	A(T22)	TEST &CUR	
00002D5C	00002730			2205+	DC	A(T23)	TEST &CUR	
00002D60	00002828			2206+	DC	A(T24)	TEST &CUR	
00002D64	00002920			2207+	DC	A(T25)	TEST &CUR	
00002D68	00002A18			2208+	DC	A(T26)	TEST &CUR	
00002D6C	00002B10			2209+	DC	A(T27)	TEST &CUR	
00002D70	00002C08			2210+	DC	A(T28)	TEST &CUR	
				2211+*				
00002D74	00000000			2212+	DC	A(0)	END OF TABLE	
00002D78	00000000			2213+	DC	A(0)		
				2214				
00002D7C	00000000			2215	DC	F'0'	END OF TABLE	
00002D80	00000000			2216	DC	F'0'		

LOC	OBJECT CODE	ADDR1	ADDR2	STMT
				2218 *****
				2219 * Register equates
				2220 *****
		00000000	00000001	2222 R0 EQU 0
		00000001	00000001	2223 R1 EQU 1
		00000002	00000001	2224 R2 EQU 2
		00000003	00000001	2225 R3 EQU 3
		00000004	00000001	2226 R4 EQU 4
		00000005	00000001	2227 R5 EQU 5
		00000006	00000001	2228 R6 EQU 6
		00000007	00000001	2229 R7 EQU 7
		00000008	00000001	2230 R8 EQU 8
		00000009	00000001	2231 R9 EQU 9
		0000000A	00000001	2232 R10 EQU 10
		0000000B	00000001	2233 R11 EQU 11
		0000000C	00000001	2234 R12 EQU 12
		0000000D	00000001	2235 R13 EQU 13
		0000000E	00000001	2236 R14 EQU 14
		0000000F	00000001	2237 R15 EQU 15
				2239 *****
				2240 * Register equates
				2241 *****
		00000000	00000001	2243 FPR0 EQU 0
		00000001	00000001	2244 FPR1 EQU 1
		00000002	00000001	2245 FPR2 EQU 2
		00000003	00000001	2246 FPR3 EQU 3
		00000004	00000001	2247 FPR4 EQU 4
		00000005	00000001	2248 FPR5 EQU 5
		00000006	00000001	2249 FPR6 EQU 6
		00000007	00000001	2250 FPR7 EQU 7
		00000008	00000001	2251 FPR8 EQU 8
		00000009	00000001	2252 FPR9 EQU 9
		0000000A	00000001	2253 FPR10 EQU 10
		0000000B	00000001	2254 FPR11 EQU 11
		0000000C	00000001	2255 FPR12 EQU 12
		0000000D	00000001	2256 FPR13 EQU 13
		0000000E	00000001	2257 FPR14 EQU 14
		0000000F	00000001	2258 FPR15 EQU 15
				2260 *****
				2261 * Register equates
				2262 *****
		00000000	00000001	2264 V0 EQU 0
		00000001	00000001	2265 V1 EQU 1
		00000002	00000001	2266 V2 EQU 2
		00000003	00000001	2267 V3 EQU 3
		00000004	00000001	2268 V4 EQU 4

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
FPCPNAME	C	00001039	8	619	378
FPCPNUM	C	00001015	3	617	376
FPC_R	F	00000040	4	721	243 248 257 278 388 424
FPC_R_1	F	00001220	4	872	891
FPC_R_10	F	00001AD8	4	1295	1314
FPC_R_11	F	00001BD0	4	1342	1361
FPC_R_12	F	00001CC8	4	1389	1408
FPC_R_13	F	00001DC0	4	1436	1455
FPC_R_14	F	00001EB8	4	1485	1504
FPC_R_15	F	00001FB0	4	1532	1551
FPC_R_16	F	000020A8	4	1579	1598
FPC_R_17	F	000021A0	4	1626	1645
FPC_R_18	F	00002298	4	1673	1692
FPC_R_19	F	00002390	4	1720	1739
FPC_R_2	F	00001318	4	919	938
FPC_R_20	F	00002488	4	1767	1786
FPC_R_21	F	00002580	4	1814	1833
FPC_R_22	F	00002678	4	1861	1880
FPC_R_23	F	00002770	4	1908	1927
FPC_R_24	F	00002868	4	1958	1977
FPC_R_25	F	00002960	4	2005	2024
FPC_R_26	F	00002A58	4	2052	2071
FPC_R_27	F	00002B50	4	2099	2118
FPC_R_28	F	00002C48	4	2146	2165
FPC_R_3	F	00001410	4	966	985
FPC_R_4	F	00001508	4	1013	1032
FPC_R_5	F	00001600	4	1060	1079
FPC_R_6	F	000016F8	4	1107	1126
FPC_R_7	F	000017F0	4	1154	1173
FPC_R_8	F	000018E8	4	1201	1220
FPC_R_9	F	000019E0	4	1248	1267
FPC_XC1	F	00000088	4	728	298 308
FPC_XC2	F	0000008C	4	729	305 311
FPC_XC_1	F	00001268	4	879	
FPC_XC_10	F	00001B20	4	1302	
FPC_XC_11	F	00001C18	4	1349	
FPC_XC_12	F	00001D10	4	1396	
FPC_XC_13	F	00001E08	4	1443	
FPC_XC_14	F	00001F00	4	1492	
FPC_XC_15	F	00001FF8	4	1539	
FPC_XC_16	F	000020F0	4	1586	
FPC_XC_17	F	000021E8	4	1633	
FPC_XC_18	F	000022E0	4	1680	
FPC_XC_19	F	000023D8	4	1727	
FPC_XC_2	F	00001360	4	926	
FPC_XC_20	F	000024D0	4	1774	
FPC_XC_21	F	000025C8	4	1821	
FPC_XC_22	F	000026C0	4	1868	
FPC_XC_23	F	000027B8	4	1915	
FPC_XC_24	F	000028B0	4	1965	
FPC_XC_25	F	000029A8	4	2012	
FPC_XC_26	F	00002AA0	4	2059	
FPC_XC_27	F	00002B98	4	2106	
FPC_XC_28	F	00002C90	4	2153	
FPC_XC_3	F	00001458	4	973	
FPC_XC_4	F	00001550	4	1020	

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES				
RPTERROR	I	00000602	4	497	350	396	432	469	
RPTSAVE	F	00000620	4	510	497	507			
RPTSVR5	F	00000624	4	511	498	506			
SKIPXC	C	00000050	8	723	275	283	288		
SKL0001	U	0000005F	1	198	214				
SKT0001	C	0000022A	26	195	198	215			
SVOLDPSW	U	00000140	0	132					
T1	A	000011E0	4	857	2183				
T10	A	00001A98	4	1280	2192				
T11	A	00001B90	4	1327	2193				
T12	A	00001C88	4	1374	2194				
T13	A	00001D80	4	1421	2195				
T14	A	00001E78	4	1470	2196				
T15	A	00001F70	4	1517	2197				
T16	A	00002068	4	1564	2198				
T17	A	00002160	4	1611	2199				
T18	A	00002258	4	1658	2200				
T19	A	00002350	4	1705	2201				
T2	A	000012D8	4	904	2184				
T20	A	00002448	4	1752	2202				
T21	A	00002540	4	1799	2203				
T22	A	00002638	4	1846	2204				
T23	A	00002730	4	1893	2205				
T24	A	00002828	4	1943	2206				
T25	A	00002920	4	1990	2207				
T26	A	00002A18	4	2037	2208				
T27	A	00002B10	4	2084	2209				
T28	A	00002C08	4	2131	2210				
T3	A	000013D0	4	951	2185				
T4	A	000014C8	4	998	2186				
T5	A	000015C0	4	1045	2187				
T6	A	000016B8	4	1092	2188				
T7	A	000017B0	4	1139	2189				
T8	A	000018A8	4	1186	2190				
T9	A	000019A0	4	1233	2191				
TESTING	F	00001004	4	607	234				
TNUM	H	00000004	2	707	233	325	372	408	445
TSUB	A	00000000	4	706	236				
TTABLE	F	00002D04	4	2182					
V0	U	00000000	1	2264					
V1	U	00000001	1	2265					
V10	U	0000000A	1	2274					
V11	U	0000000B	1	2275					
V12	U	0000000C	1	2276					
V13	U	0000000D	1	2277					
V14	U	0000000E	1	2278					
V15	U	0000000F	1	2279	297	299	304	306	
V16	U	00000010	1	2280	296	297	303	304	
V17	U	00000011	1	2281					
V18	U	00000012	1	2282					
V19	U	00000013	1	2283					
V1FUDGE	X	000011B0	16	697					
V1INPUT	X	000011C0	16	698					
V101	X	00001208	16	870	892				
V1010	X	00001AC0	16	1293	1315				
V1011	X	00001BB8	16	1340	1362				

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES	
V2_17	A	0000216C	4	1619	1642	
V2_18	A	00002264	4	1666	1689	
V2_19	A	0000235C	4	1713	1736	
V2_2	A	000012E4	4	912	935	
V2_20	A	00002454	4	1760	1783	
V2_21	A	0000254C	4	1807	1830	
V2_22	A	00002644	4	1854	1877	
V2_23	A	0000273C	4	1901	1924	
V2_24	A	00002834	4	1951	1974	
V2_25	A	0000292C	4	1998	2021	
V2_26	A	00002A24	4	2045	2068	
V2_27	A	00002B1C	4	2092	2115	
V2_28	A	00002C14	4	2139	2162	
V2_3	A	000013DC	4	959	982	
V2_4	A	000014D4	4	1006	1029	
V2_5	A	000015CC	4	1053	1076	
V2_6	A	000016C4	4	1100	1123	
V2_7	A	000017BC	4	1147	1170	
V2_8	A	000018B4	4	1194	1217	
V2_9	A	000019AC	4	1241	1264	
V3	U	00000003	1	2267		
V30	U	0000001E	1	2294		
V31	U	0000001F	1	2295		
V4	U	00000004	1	2268		
V5	U	00000005	1	2269		
V6	U	00000006	1	2270		
V7	U	00000007	1	2271		
V8	U	00000008	1	2272		
V9	U	00000009	1	2273		
VXC	X	0000000B	1	713	248	417
VXC1	X	000011EB	1	864		
VXC10	X	00001AA3	1	1287		
VXC11	X	00001B9B	1	1334		
VXC12	X	00001C93	1	1381		
VXC13	X	00001D8B	1	1428		
VXC14	X	00001E83	1	1477		
VXC15	X	00001F7B	1	1524		
VXC16	X	00002073	1	1571		
VXC17	X	0000216B	1	1618		
VXC18	X	00002263	1	1665		
VXC19	X	0000235B	1	1712		
VXC2	X	000012E3	1	911		
VXC20	X	00002453	1	1759		
VXC21	X	0000254B	1	1806		
VXC22	X	00002643	1	1853		
VXC23	X	0000273B	1	1900		
VXC24	X	00002833	1	1950		
VXC25	X	0000292B	1	1997		
VXC26	X	00002A23	1	2044		
VXC27	X	00002B1B	1	2091		
VXC28	X	00002C13	1	2138		
VXC3	X	000013DB	1	958		
VXC4	X	000014D3	1	1005		
VXC5	X	000015CB	1	1052		
VXC6	X	000016C3	1	1099		
VXC7	X	000017BB	1	1146		

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES	
VXC8	X	000018B3	1	1193		
VXC9	X	000019AB	1	1240		
VXCMSG	U	00000502	1	404	250	
VXCPEXP	C	000010B5	3	640	421	
VXCPGOT	C	000010CA	3	643	428	
VXCPLINE	C	0000106B	13	635	645	431
VXCPLNG	U	00000063	1	645	430	
VXCPNAME	C	0000109C	8	638	414	
VXCPNUM	C	00001078	3	636	412	
WK1	F	00000098	4	731		
WK2	F	0000009C	4	732		
X0001	U	000002C0	1	204	192	205
X1	F	00001280	4	885	857	
X10	F	00001B38	4	1308	1280	
X11	F	00001C30	4	1355	1327	
X12	F	00001D28	4	1402	1374	
X13	F	00001E20	4	1449	1421	
X14	F	00001F18	4	1498	1470	
X15	F	00002010	4	1545	1517	
X16	F	00002108	4	1592	1564	
X17	F	00002200	4	1639	1611	
X18	F	000022F8	4	1686	1658	
X19	F	000023F0	4	1733	1705	
X2	F	00001378	4	932	904	
X20	F	000024E8	4	1780	1752	
X21	F	000025E0	4	1827	1799	
X22	F	000026D8	4	1874	1846	
X23	F	000027D0	4	1921	1893	
X24	F	000028C8	4	1971	1943	
X25	F	000029C0	4	2018	1990	
X26	F	00002AB8	4	2065	2037	
X27	F	00002BB0	4	2112	2084	
X28	F	00002CA8	4	2159	2131	
X3	F	00001470	4	979	951	
X4	F	00001568	4	1026	998	
X5	F	00001660	4	1073	1045	
X6	F	00001758	4	1120	1092	
X7	F	00001850	4	1167	1139	
X8	F	00001948	4	1214	1186	
X9	F	00001A40	4	1261	1233	
XC0001	U	000002E8	1	218	210	
XCFAILMSG	H	000003D6	2	321	316	
XCHECK	U	0000034E	1	274	239	
XC01	X	00001240	16	876		
XC010	X	00001AF8	16	1299		
XC011	X	00001BF0	16	1346		
XC012	X	00001CE8	16	1393		
XC013	X	00001DE0	16	1440		
XC014	X	00001ED8	16	1489		
XC015	X	00001FD0	16	1536		
XC016	X	000020C8	16	1583		
XC017	X	000021C0	16	1630		
XC018	X	000022B8	16	1677		
XC019	X	000023B0	16	1724		
XC02	X	00001338	16	923		
XC020	X	000024A8	16	1771		

DESC	SYMBOL	SIZE	POS	ADDR
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Entry: 0

Image	IMAGE	11652	0000-2D83	0000-2D83
Region		11652	0000-2D83	0000-2D83
CSECT	ZVE6TST	11652	0000-2D83	0000-2D83

STMT

FILE NAME

```
1 /devstor/sharedvfp/tests/zvector-e6-21-VCRNF.asm
```

**** NO ERRORS FOUND ****